

# EE104 Fundamentals of Electric Circuits (Lesson 08)

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# Important updates

- Important updates

**Chapters:** chapters 1-7      **Time:** 16:20 – 18:20 Nov.9<sup>th</sup> Saturday      **Exam:** closed-book in English.

**Location:** The 3<sup>rd</sup> teaching building, and rooms can be found in the table below.

**Tools:** answer sheets and stretch paper will be provided. You will need calculators, pens and student ID cards. No phones, no smart watches, no books or any other documents related to circuits.

**Key info. 1:** One A4 cheating paper is allowed (both side, hand writing or printing).

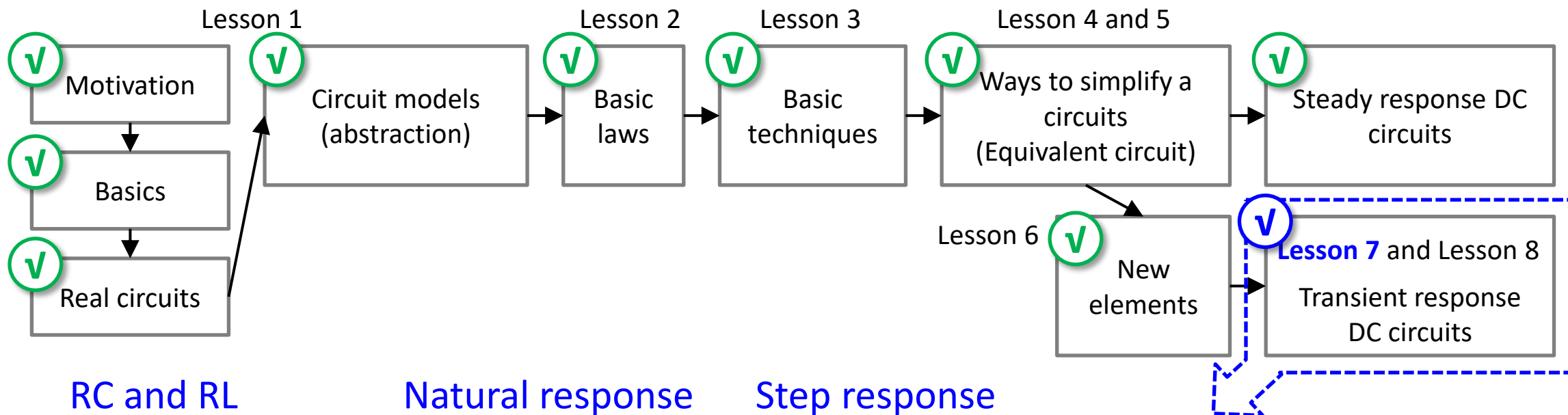
**Key info. 2:** No make-up or online exams.

**Key info. 3:** SUBMIT all the exam paper, answer sheets, scratch papers, your cheating paper. **IT IS YOUR DUTY** to make sure all these documents are safely submitted to the TA at the end of the exam. IF a student loses one of these documents, he will be considered as **CHEATING** and score will be **ZERO**.

| Class | 1     | 2       | 3       | 4       |
|-------|-------|---------|---------|---------|
| Rooms | Rm105 | Rm. 106 | Rm. 107 | Rm. 108 |
| 教室    | 三教105 | 三教106   | 三教107   | 三教108   |

# Review of Lesson 7(a)

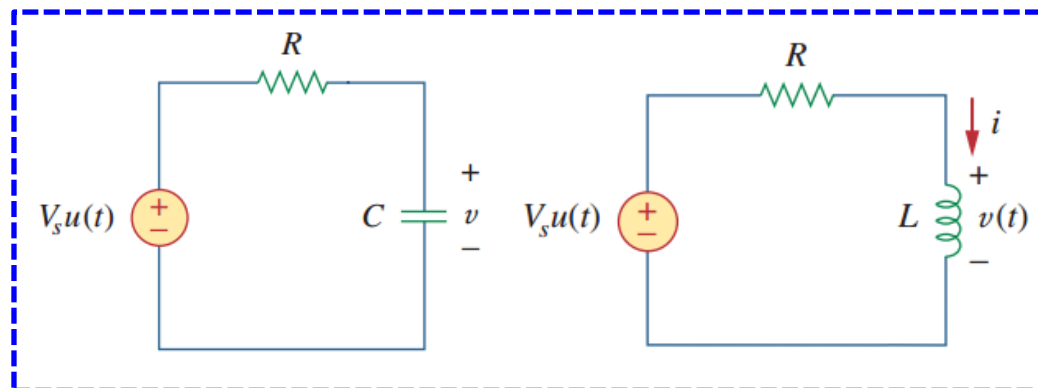
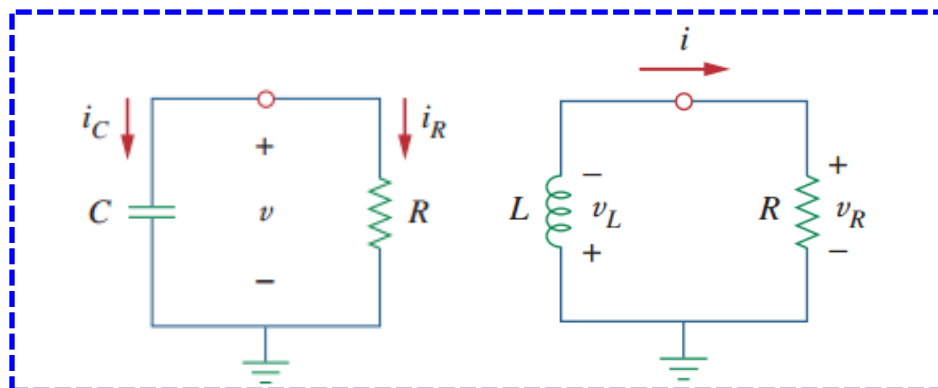
- EE104 road map



RC and RL

Natural response

Step response



# Review of Lesson 7(b)

- Source-free RC and RL circuits (Natural response):

RC

$$C \frac{dv}{dt} + \frac{v}{R} = 0 \rightarrow \frac{dv}{dt} + \frac{v}{RC} = 0$$

$$v(t) = V_0 e^{-t/RC} = V_0 e^{-t/\tau}$$

1. The **initial voltage**  $v(t) = V_0$  on capacitor. The initial voltage can be obtained from the steady analysis of the circuit (treat capacitor as open circuit)
2. The **time constant**  $\tau = RC$ . In finding the time constant  $\tau$ , one may need to use the Thevenin theorem to find out the equivalent resistance seen by the capacitor (the circuit seen by the capacitor may contain more than one resistors and dependent sources).

RL

$$L \frac{di}{dt} + Ri = 0 \rightarrow \frac{di}{dt} + \frac{R}{L} i = 0$$

$$i(t) = i(0) e^{-Rt/L} = i(0) e^{-t/\tau}$$

1. The **initial current**  $i(t) = i_0$  through the inductor. The initial current can be obtained from the steady analysis of the circuit (treat capacitor as short circuit).
2. The **time constant**  $\tau = L/R$ . In finding the time constant  $\tau$ , one may need to use the Thevenin theorem to find out the equivalent resistance seen by the inductor (the circuit seen by the capacitor may contain more than one resistors and dependent sources).

# Review of Lesson 7(c)

- Step responses of RC and RL circuits:

$$v(t) = v_{ss} + v_t = v(\infty) + [v(0) - v(\infty)]e^{-t/\tau}$$

- a) Find - **the initial capacitor voltage  $v(0)$**   
- **the final capacitor voltage  $v(\infty)$**   
- **the time constant  $\tau = RC$ .**

- b) If  $t_0 \neq 0$

$$v(t) = v(\infty) + [v(t_0) - v(\infty)]e^{-(t-t_0)/\tau}$$

$$i(t) = i_{ss} + i_t = i(\infty) + [i(0) - i(\infty)]e^{-t/\tau}$$

- a) Find - **the initial inductor current  $i(0)$**   
- **the final inductor current  $v(\infty)$**   
- **the time constant  $\tau = L/R$ .**

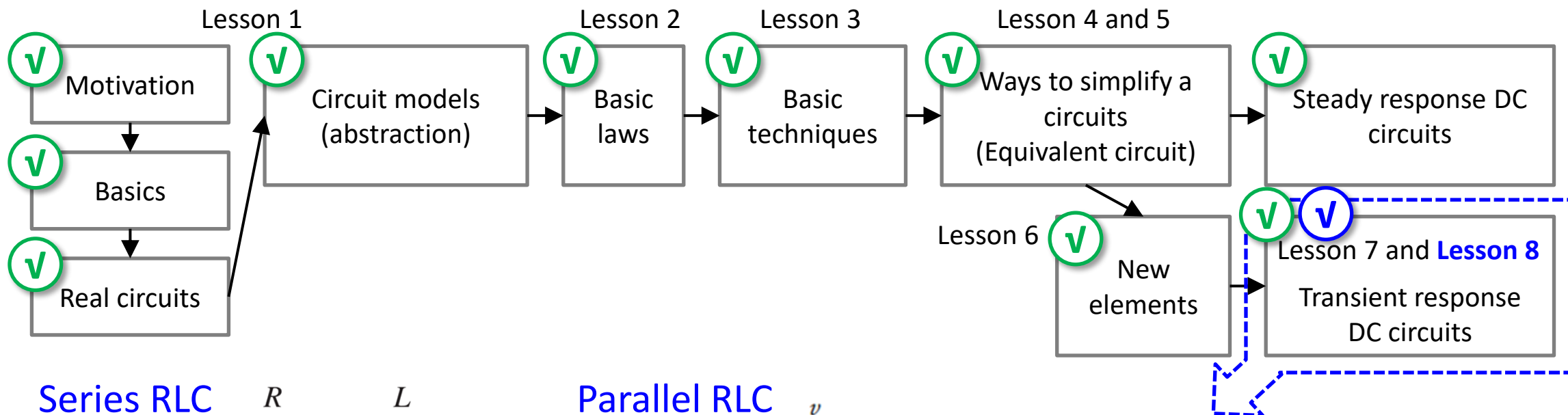
- b) If  $t_0 \neq 0$

$$i(t) = i(\infty) + [i(t_0) - i(\infty)]e^{-(t-t_0)/\tau}$$

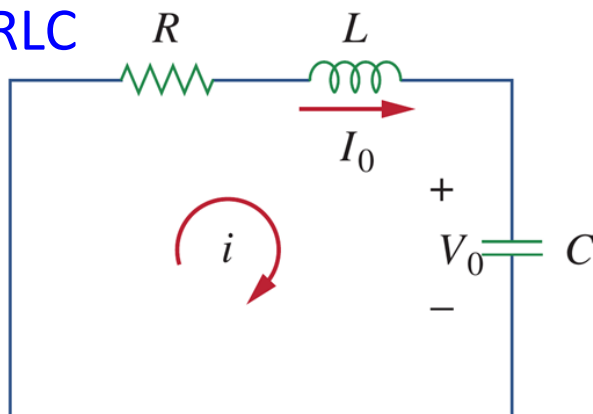


# Abstract of Lesson 8

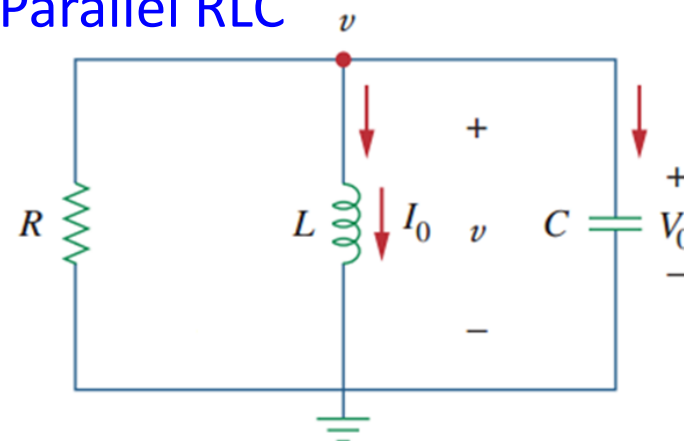
- EE104 road map



Series RLC



Parallel RLC



Natural response  
Step response

# Lesson 8 Second-order circuits

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| <b>8.1-8.3</b> | <b>Second-order circuits</b> | <b>Slides 08-20</b> |
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| 8.4-8.6 | Natural response of RLC circuits | Slides 22-50 |
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| 8.7-8.8 | Step response of RLC circuits | Slides 52-70 |
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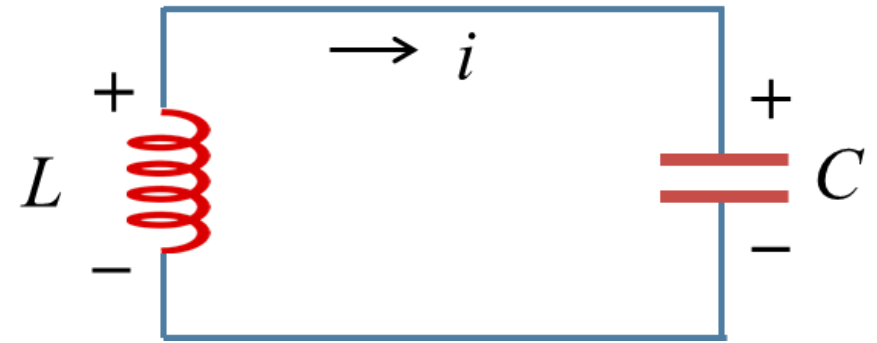
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| 8.9 | Summary | Slides 72-73 |
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## 8.1 Second-order circuits (a)

- Second-order circuits

- In the previous chapter we considered circuits with a single storage element (a capacitor or an inductor), which are first-order circuits.
- In this chapter we will consider circuits containing two storage elements. These are known as **second-order circuits (二阶电路)** because their responses are described by differential equations that contain second derivatives.



$$v_C = v_L = L \frac{di}{dt} \quad \& \quad i_L = i_C = i = C \frac{dv_C}{dt}$$

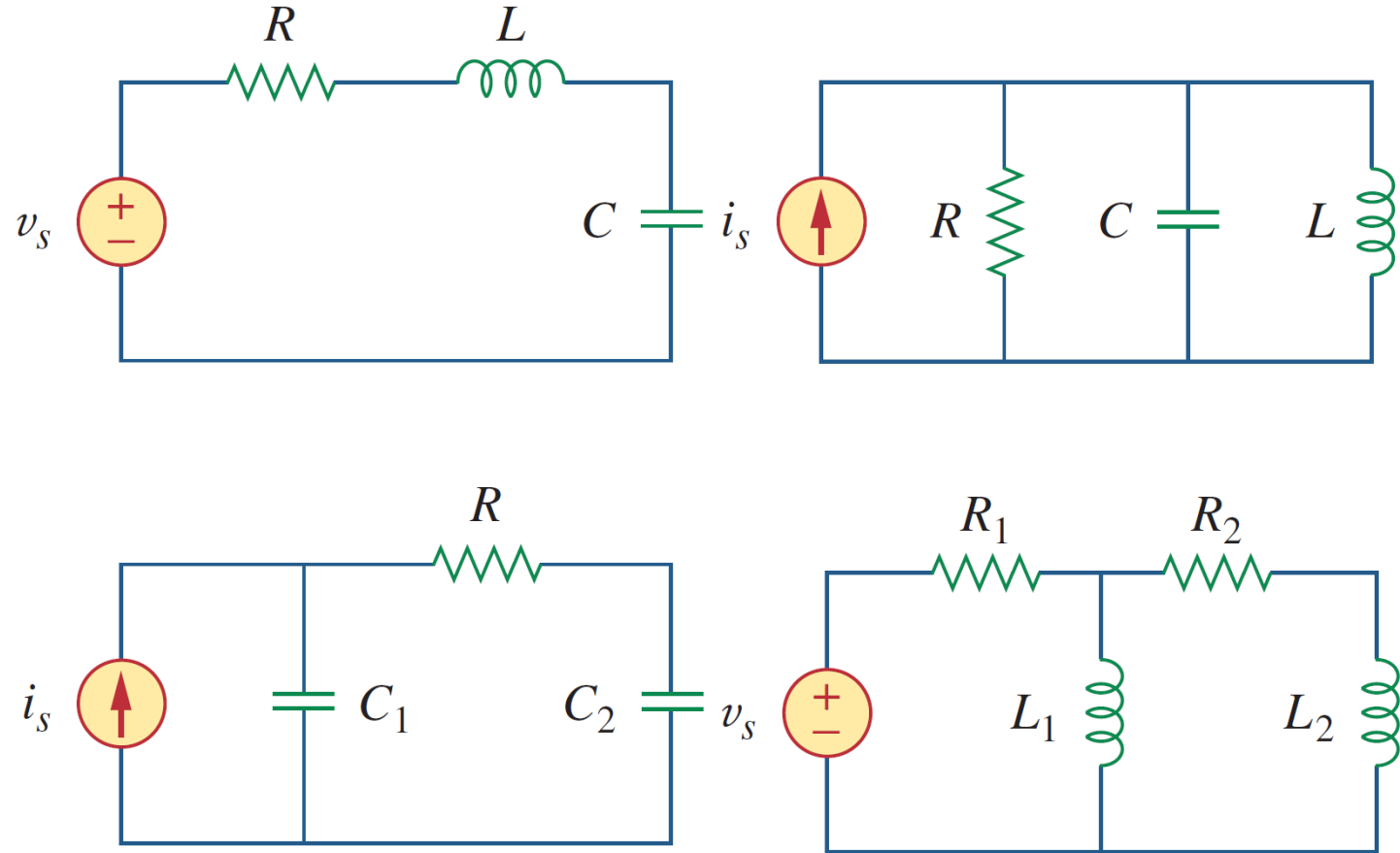
$$\Rightarrow i = LC \frac{d^2 i}{dt^2} \quad \Rightarrow \frac{d^2 i}{dt^2} + \frac{1}{LC} i = 0$$



## 8.1 Second-order circuits (b)

- Second-order circuits

- a) Note that the two storage elements can be of different type (a and b) or the same type (c and d). Other components in the circuit can be op amps, resistors, dependent and/or independent sources.
- b) The circuits can either be excited from (1) initial storage energy (natural response) and (2) by switching on independent sources (independent). We will study the response in both scenarios.



## 8.2 Second-order DEs (a)

- Homogeneous second-order equations

$$ay'' + by' + cy = 0$$

- a) Assume  $e^{rx}$  is the solution (taking advantage of that the derivative of an exponential function is also an exponential function of the same form).
- b) The equation turns into:

$$\begin{aligned} ar^2e^{rx} + bre^{rx} + ce^{rx} &= 0 \Rightarrow (ar^2 + br + c)e^{rx} = 0 \\ &\Rightarrow ar^2 + br + c = 0 \end{aligned}$$

- c)  $ar^2 + br + c = 0$  is called the characteristic equation (特征方程)
- d) Depending on the roots, the general solution of equation [1] can be different.

## 8.2 Second-order DEs (b)

- Homogeneous second-order equations

$$ar^2 + br + c = 0 \quad [1]$$

$$\downarrow$$
$$r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$
$$\downarrow$$

$$ay'' + by' + cy = 0 \quad [2]$$

$$\downarrow$$
$$y = e^{rx}$$
$$\downarrow$$

Roots of  $ar^2 + br + c = 0$

$r_1, r_2$  real and distinct

$r_1 = r_2 = r$

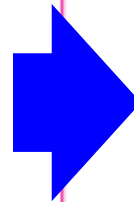
$r_1, r_2$  complex:  $\alpha \pm i\beta$

General solution

$$y = c_1 e^{r_1 x} + c_2 e^{r_2 x}$$

$$y = c_1 e^{rx} + c_2 x e^{rx}$$

$$y = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$$



## 8.2 Second-order DEs (c)

- Inhomogeneous second-order equations

$$ay'' + by' + cy = g(x)$$

- a)  $a, b, c$  are constants and  $g(x)$  is a continuous function, the solution can be written as the sum of the common solution,  $y_c(x)$ , and a particular solution of the inhomogeneous  $y_p(x)$ , i.e.,  $y(x) = y_c(x) + y_p(x)$ .
- a) The particular solution can be found by the method of undetermined coefficients (待定系数法) for the following cases relevant to EE104.
- When  $G(x)$  is a polynomial of degree  $n$  ( $n$  阶多项式),  $y_p(x)$  is a polynomial of the same order.
  - When  $G(x) = Ce^{kx}$  where  $C$  and  $k$  are constants,  $y_p(x) = Ae^{kx}$ , where  $A$  is a constant to be determined.
  - When  $G(x) = C \cdot \cos(x)$  or  $C \cdot \sin(x)$ ,  $y_p(x) = A \cdot \cos(x) + B \cdot \sin(x)$ .

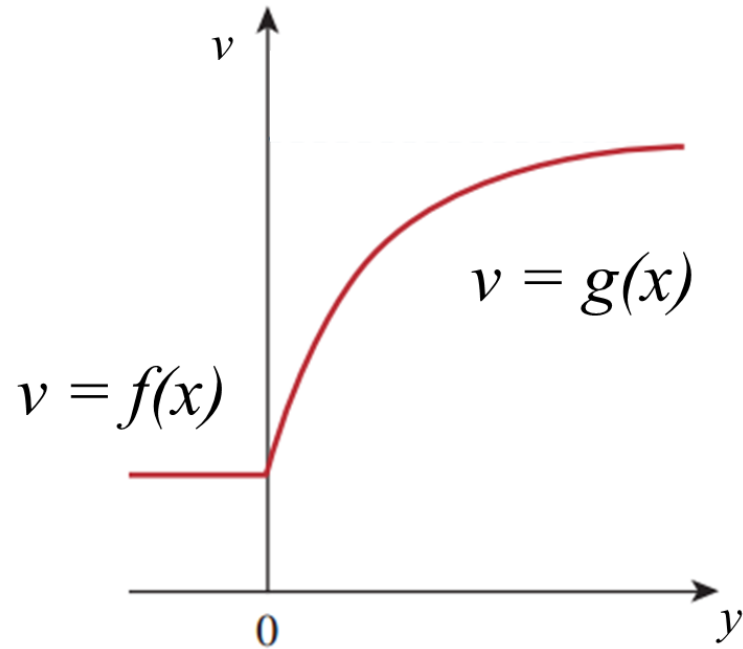
## 8.3 Initial values (a)

- Initial values

- a) In resolving the first-order circuits, we need  $v(0)$  or  $i(0)$  to help us determine the natural response or the transient response.
- b) To resolve the second-order circuits, we also need  $v'(0)$  or  $i'(0)$ .
- c) As always in circuit analysis, we must carefully handle the polarity of voltage across a capacitor and the direction of current through an inductor so that they conform to the passive sign convention.
- d) Capacitor voltage and inductor current always remain continuous. Therefore,  $v(0^+) = v(0^-)$  for capacitor, and  $i(0^+) = i(0^-)$  for inductor, where  $t = 0^+$ ,  $0^-$  denotes the time just before and just after the starting instant or the moment that switching event occurs.

## 8.3 Initial values (b)

- $v(0^+)$  and  $v(0^-)$  can be roughly considered as follows



$$\begin{cases} v(0^+) = g(0) \\ v(0^-) = f(0) \end{cases} \quad \begin{cases} v'(0^+) = g'(0) \\ v'(0^-) = f'(0) \end{cases}$$

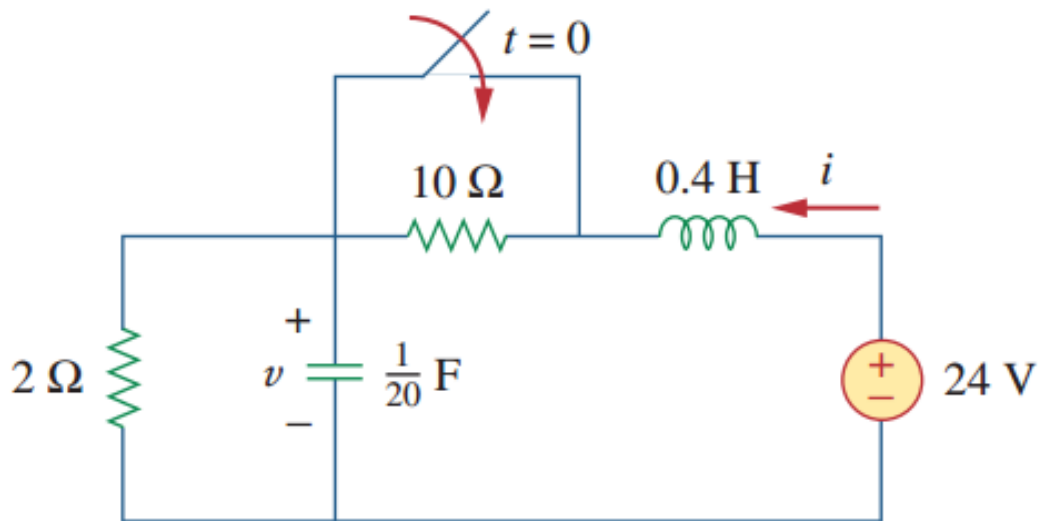
- For capacitors,  $v(0^+) = v(0^-)$  since the voltage across a capacitor cannot change abruptly.
- However,  $v'(0^+) \neq v'(0^-)$ , so  $i_c(0^+) \neq i_c(0^-)$
- For simplicity,  $v(0^+)$  and  $v'(0^+)$  are written as  $v(0)$  and  $v'(0)$  in the following slides.
- Think: is  $v(0^+) = v(0^-)$  true for resistors inductors?
- Think: how about  $v(\infty)$  and  $v'(\infty)$  for capacitors?
- In following slides,  $v$  represents the voltage across the capacitor and  $i$  represents the current through an inductor if not specified.



## 8.3 Initial values (c)

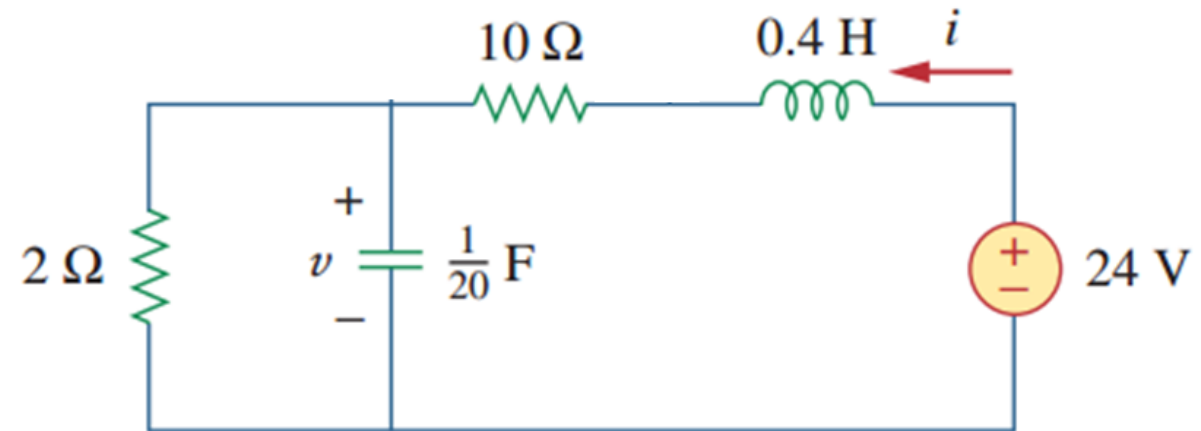
### • Example 8.01

**Example 8.01:** The switch has been open for a long time and was closed at  $t = 0$ . Determine: (a)  $i(0)$ ,  $v(0)$ ; (b)  $i'(0)$ ,  $v'(0)$ , and (c)  $i(\infty)$ ,  $v(\infty)$ .



**Solution:**

(1) Find the **initial values** using the equivalent circuit before the stimulus.



The capacitor is like an open circuit:

$$v(0) = 4 \text{ V}$$

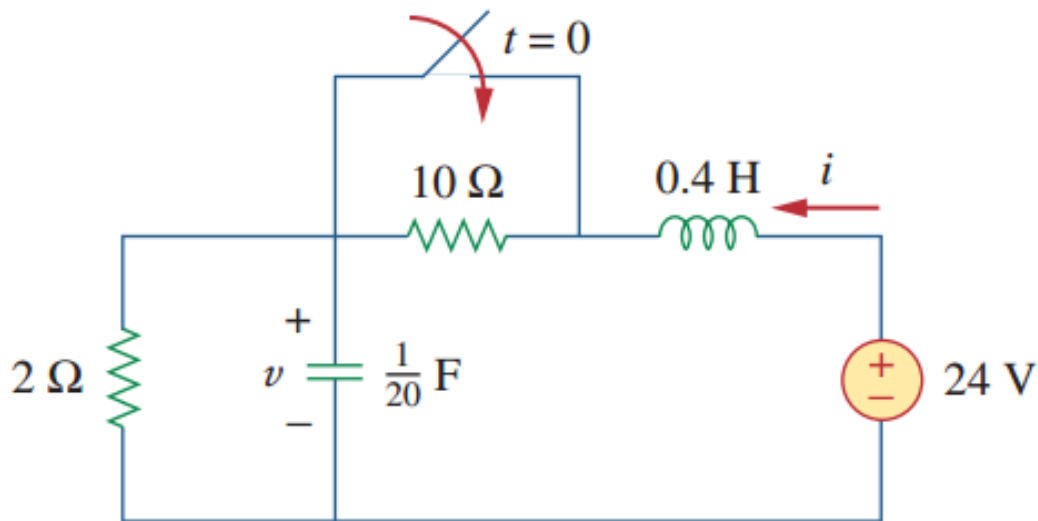
The inductor is like an ideal wire:

$$i(0) = 24/12 = 2 \text{ A}$$

## 8.3 Initial values (d)

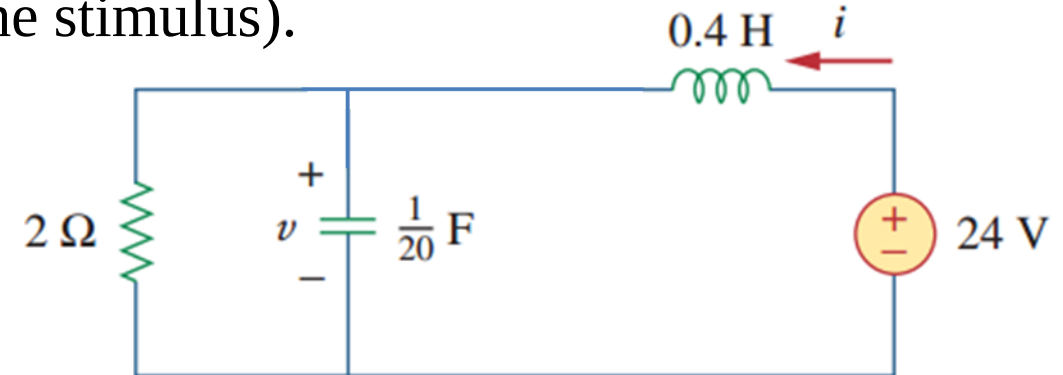
### • Example 8.01

**Example 8.01:** The switch has been open for a long time and was closed at  $t = 0$ . Determine: (a)  $i(0)$ ,  $v(0)$ ; (b)  $i'(0)$ ,  $v'(0)$ , and (c)  $i(\infty)$ ,  $v(\infty)$ .



**Solution:**

(2) Find the **initial derivatives** using the equivalent circuit after the stimulus (just after the stimulus).



KVL:  $24 = i'(0)L + v(0)$

$$i'(0) = [24 - v(0)]/L = 50 \text{ A/s}$$

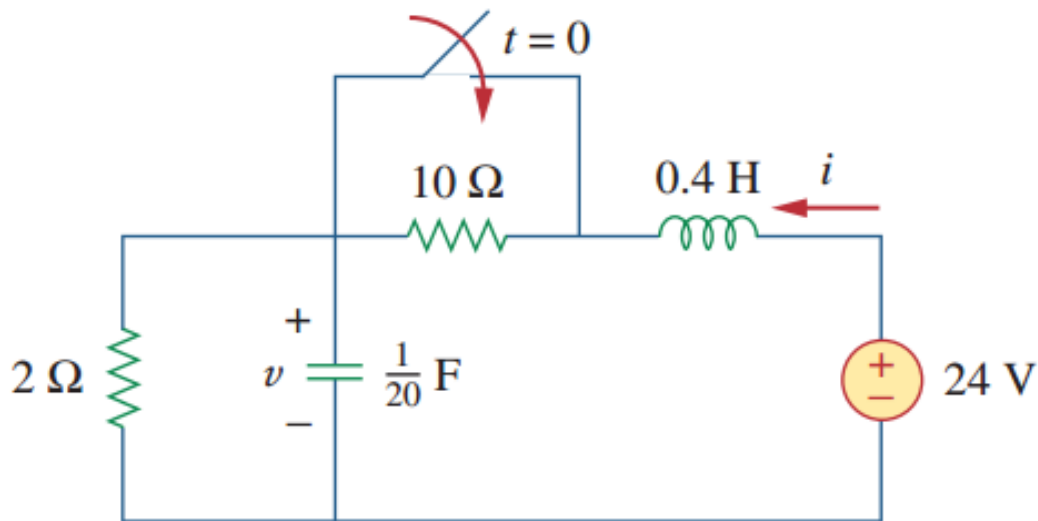
KCL:  $i(0) = v'(0)C + v(0)/2$

$$v'(0) = i(0) - v(0)/2 = 0 \text{ V/s}$$

## 8.3 Initial values (e)

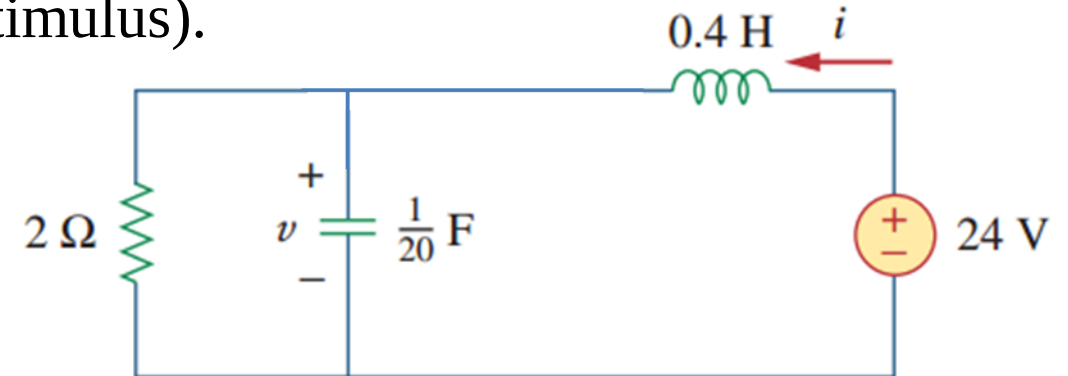
### • Example 8.01

**Example 8.01:** The switch has been open for a long time and was closed at  $t = 0$ . Determine: (a)  $i(0)$ ,  $v(0)$ ; (b)  $i'(0)$ ,  $v'(0)$ , and (c)  $i(\infty)$ ,  $v(\infty)$ .



**Solution:**

(3) Find the **final values** using the equivalent circuit after the stimulus (long time after the stimulus).



The capacitor is like an open circuit,:

$$v(\infty) = 24 \text{ V}$$

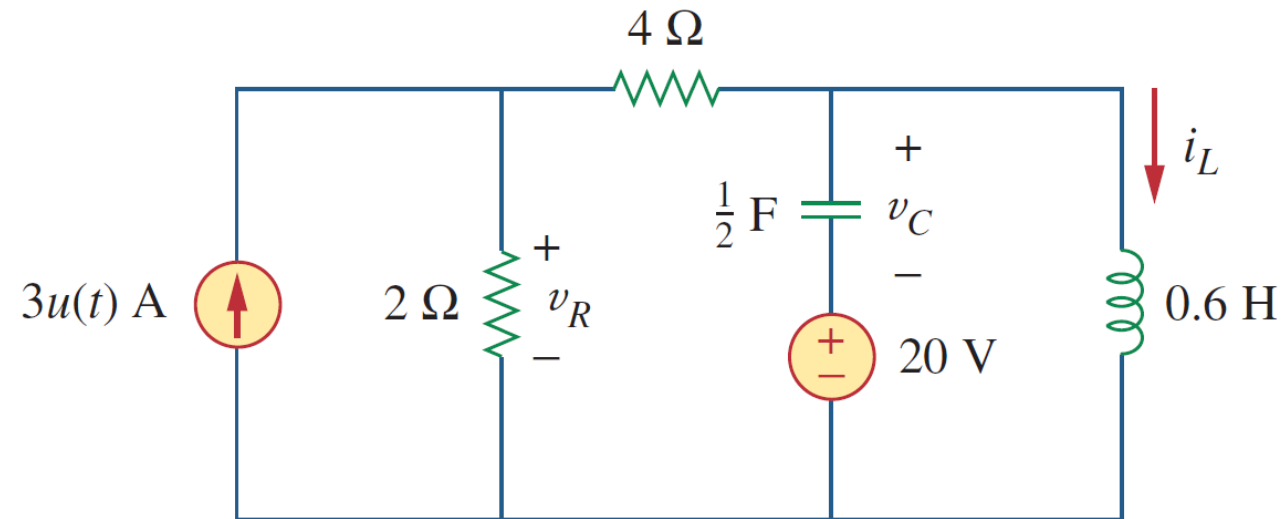
The inductor is like an ideal wire:

$$i(\infty) = 24/2 = 12 \text{ A}$$

## 8.3 Initial values (f)

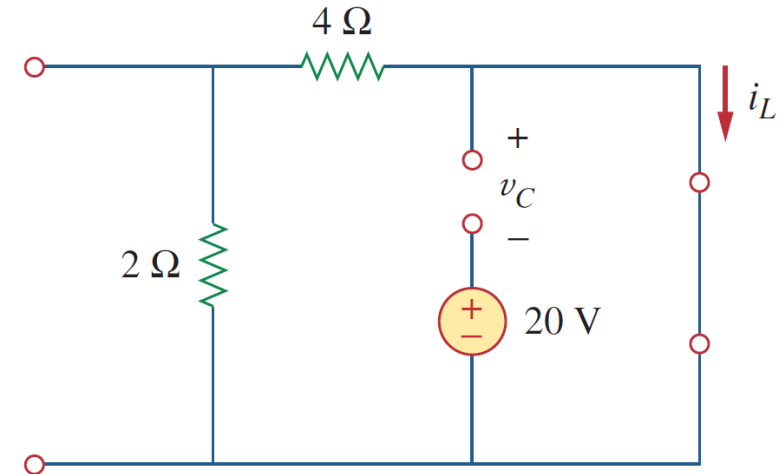
- Example 8.02

Example 8.02: Determine: (a)  $i(0)$ ,  $v(0)$ , (b)  $v'(0)$ ,  $i'(0)$ , (c)  $v(\infty)$ ,  $i(\infty)$



Solution:

(1) Find the **initial values** using the equivalent circuit before the stimulus.



The capacitor is like an open circuit:  
 $v_C(0) = -20 \text{ V}$ ,  $v_{R4}(0) = 0 \text{ V}$ ,  $v_{R2}(0) = 0 \text{ V}$

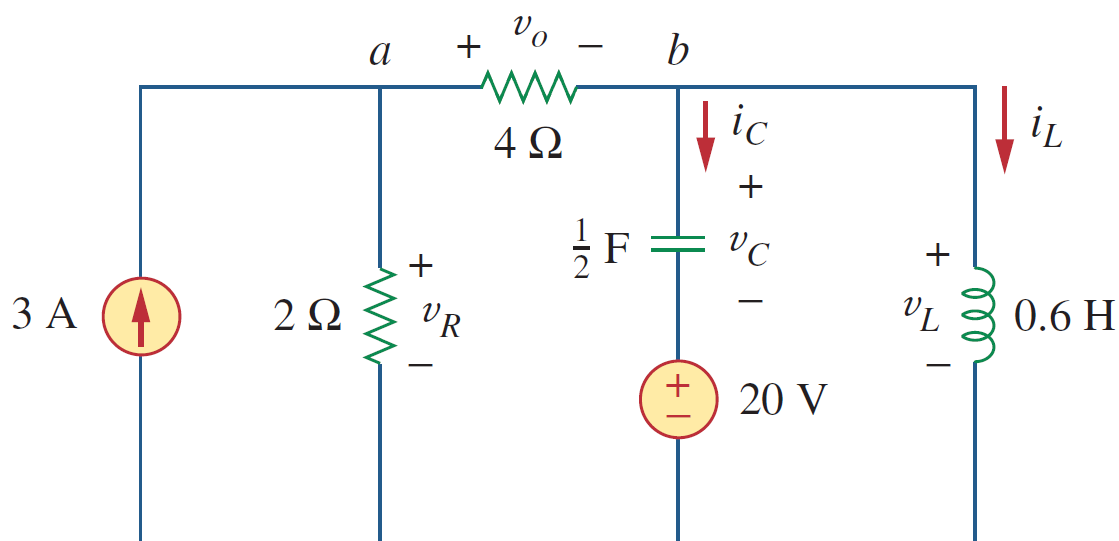
The inductor is like an ideal wire:

$$i(0) = 0 \text{ A}$$

## 8.3 Initial values (g)

### • Example 8.02

**Example 8.02:** Determine: (a)  $i(0)$ ,  $v(0)$ , (b)  $v'(0)$ ,  $i'(0)$ , (c)  $v(\infty)$ ,  $i(\infty)$



**Solution:**

(2) Find the **initial derivatives** using the equivalent circuit after the stimulus (just after the stimulus).

$$v_C(0^+) = v_C(0^-) = -20 \text{ V}, i_L(0^+) = i_L(0^-) = 0 \text{ A}$$

$$\begin{cases} \frac{v_{R4}(0^+)}{4} + \frac{v_{R2}(0^+)}{2} = 3 \text{ A} \\ \frac{v_{R4}(0^+)}{4} = i_C(0^+) + 0 \\ -v_{R2}(0^+) + v_{R4}(0^+) + v_{R4}(0^+) + 20 = 0 \end{cases}$$

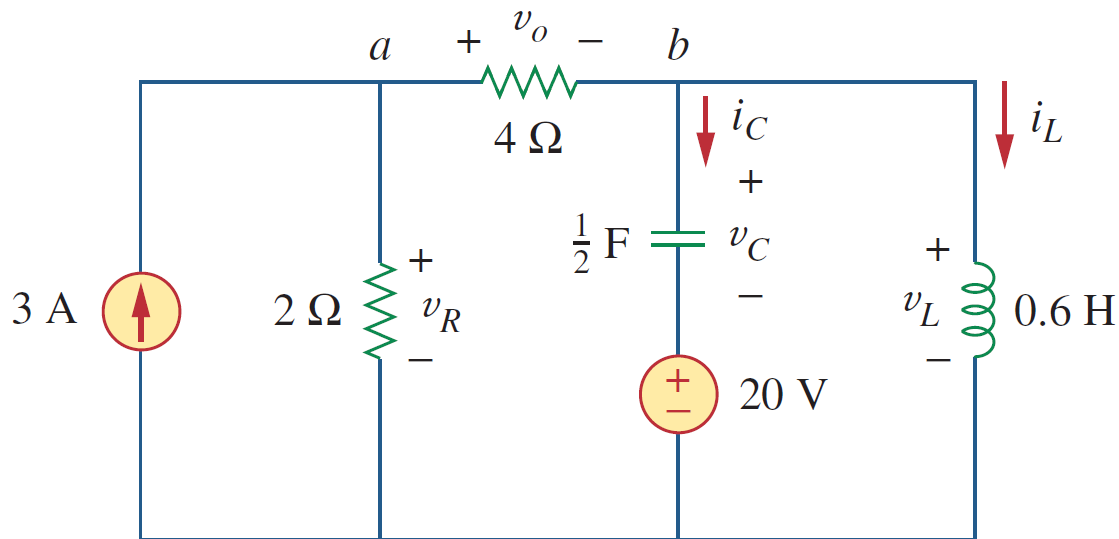
$$\Rightarrow v_{R4}(0^+) = 4 \text{ V} \Rightarrow i_C(0^+) = 1 \text{ A}$$

$$\Rightarrow v'(0) = i_C(0^+)/C = 2 \text{ V/s}$$

## 8.3 Initial values (h)

### • Example 8.02

**Example 8.02:** Determine: (a)  $i(0)$ ,  $v(0)$ , (b)  $v'(0)$ ,  $i'(0)$ , (c)  $v(\infty)$ ,  $i(\infty)$



**Solution:**

(2) Find the **initial derivatives** using the equivalent circuit after the stimulus (just after the stimulus).

Using KVL for the right mesh:

$$-20 - v_C(0) + v_L(0^+) = 0 \Rightarrow v_L(0^+) = 0$$

$$\text{Therefore } i'(0) = v_L(0^+)/L = 0 \text{ A/s}$$

(3) Find the **final values** using the equivalent circuit after the stimulus (long time after the stimulus).

$$v(\infty) = -20 \text{ V}$$

$$i(\infty) = 24/2 = 1 \text{ A}$$

# Lesson 8 Second-order circuits

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| 8.1-8.3 | Second-order circuits | Slides 08-20 |
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| <b>8.4-8.6</b> | <b>Natural response of RLC circuits</b> | <b>Slides 22-50</b> |
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| 8.7-8.8 | Step response of RLC circuits | Slides 52-70 |
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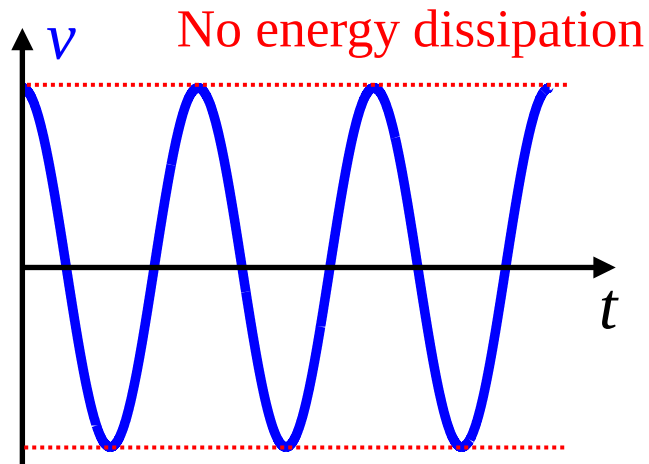
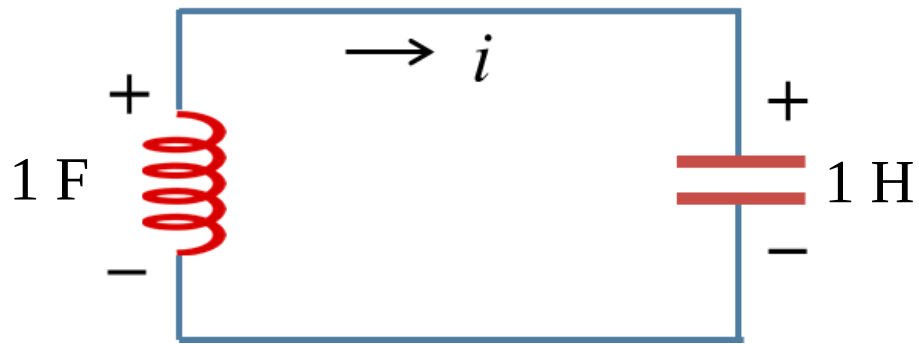
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| 8.9 | Summary | Slides 72-73 |
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## 8.4 Introduction: LC circuits and oscillation (a)

- Look at the following circuit



The circuit is based on ideal elements. The initial voltage  $v(0)$  is 1 V and the initial current  $i(0)$  is 0 A at  $t = 0$  S.

$$v = v_C = v_L = L \frac{di}{dt} \quad \& \quad i = i_L = i_C = C \frac{dv_C}{dt}$$

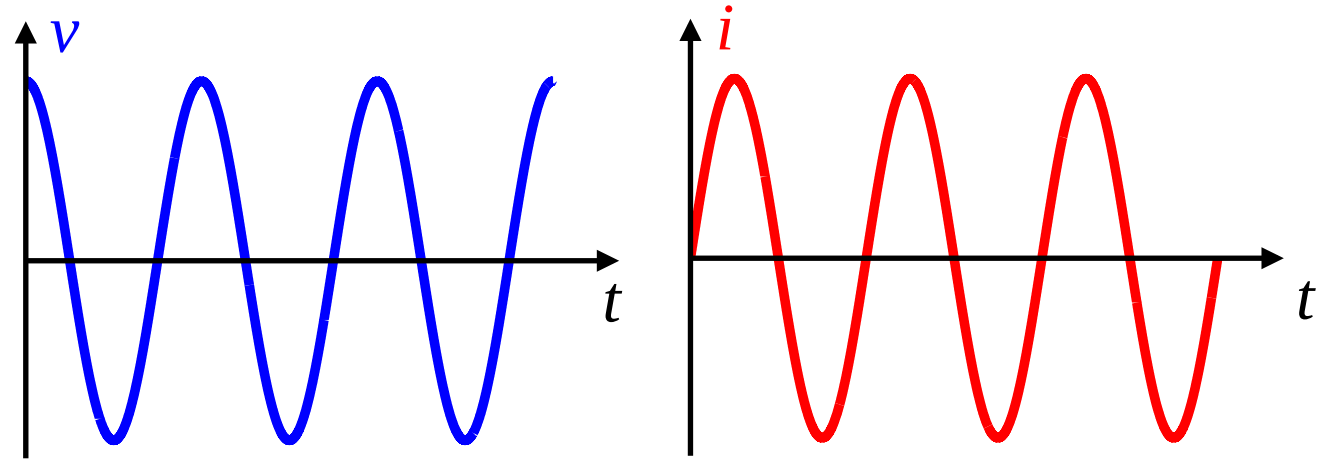
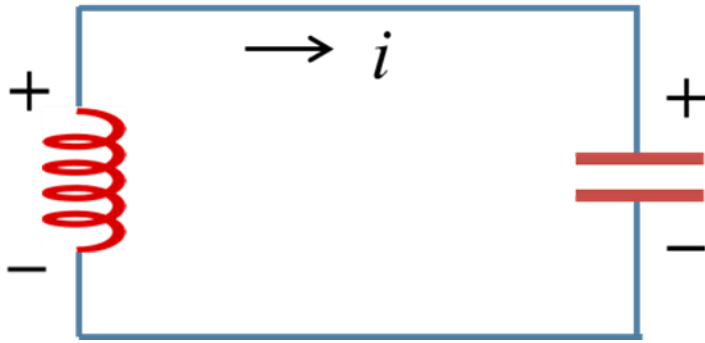
$$\Rightarrow i = LC \frac{d^2 i}{dt^2} \quad \Rightarrow \frac{d^2 i}{dt^2} + \frac{1}{LC} i = 0$$

$$\Rightarrow v = \cos t \quad \& \quad i = \sin t$$

$$\Rightarrow w = \frac{1}{2} C v^2 + \frac{1}{2} L i^2 = 1 \text{ (常数)}$$

## 8.4 Introduction: LC circuits and oscillation (b)

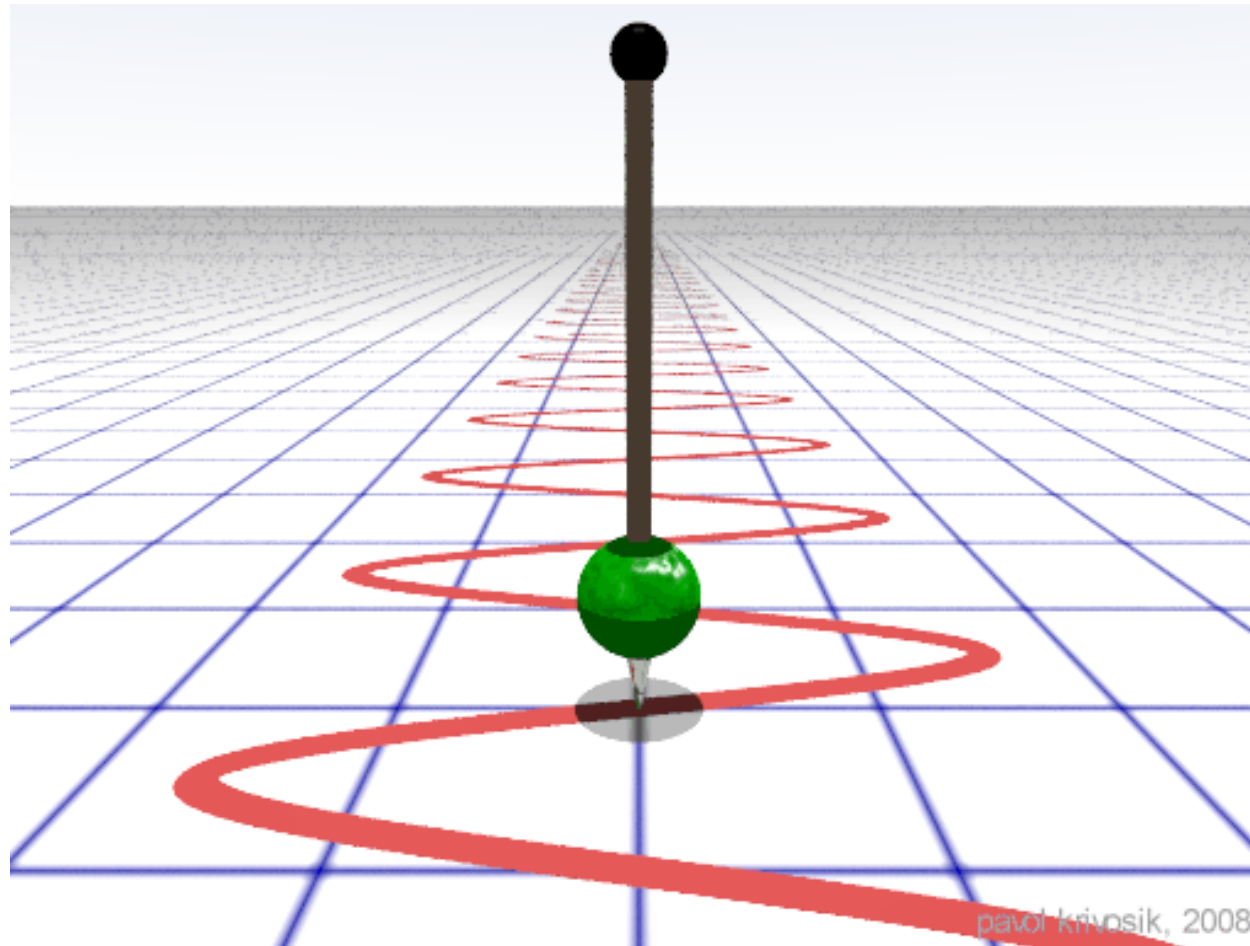
- Electrical oscillation



- Electrical **oscillation** (震荡) is a repeating cyclical variation in voltage or current in an electrical circuit, resulting in a periodic waveform.
- A parallel  $LC$  circuit, which is also called as tank or oscillatory circuit, produces oscillations at the **resonance frequency** (谐振频率)  $\omega_0 = 1/\sqrt{LC}$ .
- No loss of energy if these are ideal elements

## 8.4 Introduction: LC circuits and oscillation (c)

- Periodic transfer of energy without dissipation



## 8.4 Introduction: LC circuits and oscillation (c)

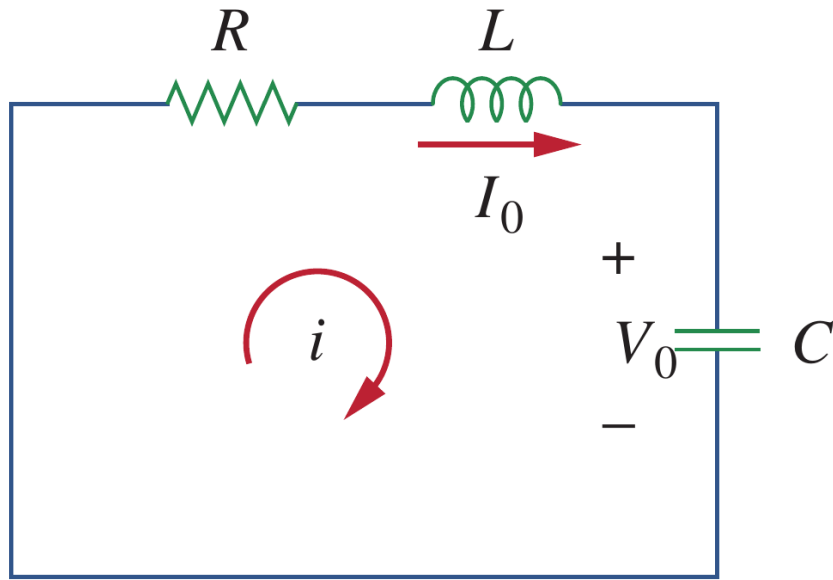
- What if a resistor is added?

Check this website: <https://falstad.com/circuit/>

Click the “full screen version”

## 8.5 Natural response of series RLC circuits (a)

- Natural response of series RCL circuits



- a) Excited from the initial stored energy: voltage across the capacitor  $V_0$  and/or current through the inductor  $I_0$ .
- b) Using KVL:

$$Ri + L \frac{di}{dt} + \frac{1}{C} \int_{-\infty}^t i(\tau) d\tau = 0$$

$$\frac{d^2 i}{dt^2} + \frac{R}{L} \frac{di}{dt} + \frac{i}{LC} = 0$$

Which is a homogenous second-order DE

## 8.5 Natural response of series RLC circuits (b)

- Resonant frequency and damping factor

The homogenous second-order DE:  $\frac{d^2 i}{dt^2} + \frac{R}{L} \frac{di}{dt} + \frac{i}{LC} = 0$

The characteristic equation:  $s^2 + \frac{R}{L}s + \frac{1}{LC} = 0$

$$\Rightarrow s_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}} \quad s_2 = -\frac{R}{2L} - \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$$

$$\Rightarrow s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}, \quad s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$$

$$\alpha = \frac{R}{2L}, \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

- a) The roots  $s_1$  and  $s_2$  are called **natural frequencies**, measured in nepers per second (Np/s).
- b)  $\omega_0$  is the **resonant frequency**, expressed in radians per second (rad/s);
- c)  $\alpha$  is the neper frequency (奈培频率) or the **damping factor** (阻尼系数), expressed in nepers per second.

## 8.5 Natural response of series RLC circuits (c)

- $\alpha > \omega_0$ : overdamped (过阻尼)

a) When  $\alpha > \omega_0$ , we have:

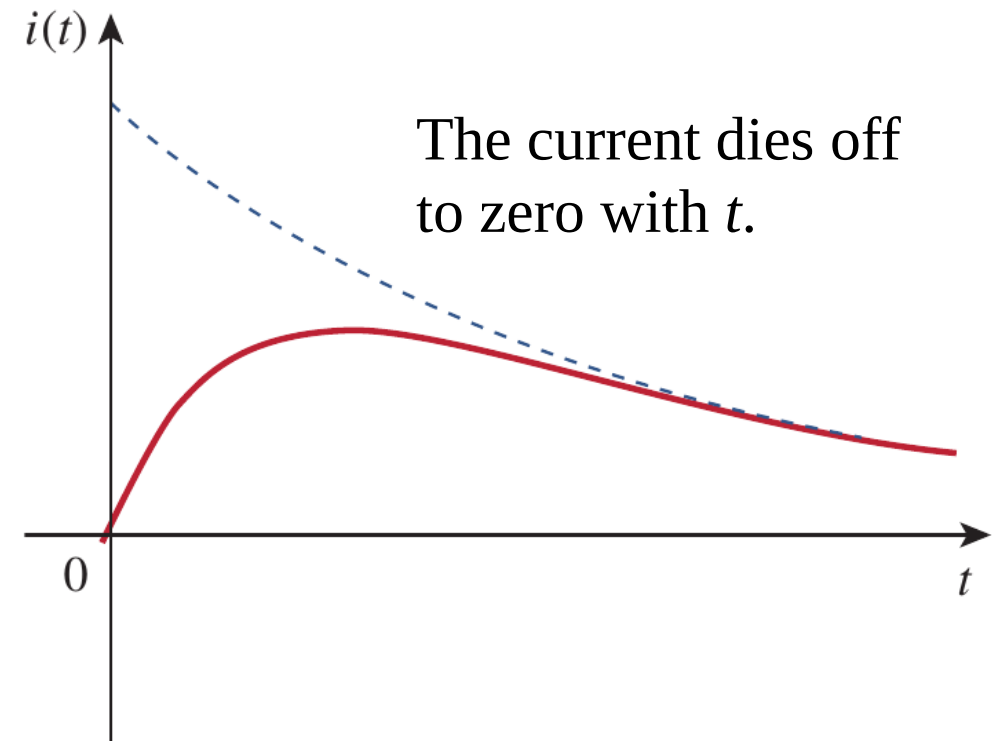
$$\left(\frac{R}{2L}\right)^2 - \frac{1}{LC} = \left(\frac{R}{2L} + \frac{1}{\sqrt{LC}}\right)\left(\frac{R}{2L} - \frac{1}{\sqrt{LC}}\right) > 0$$

b) Therefore  $s_1$  and  $s_2$  are different negative real numbers and the solution is:

$$i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

c) You can use initial values to find  $A_1$  and  $A_2$

$$\alpha = \frac{R}{2L}, \quad \omega_0 = \frac{1}{\sqrt{LC}}$$



## 8.5 Natural response of series RLC circuits (d)

- $\alpha = \omega_0$ : critically damped (临界阻尼)

a) When  $\alpha = \omega_0$ , we have:

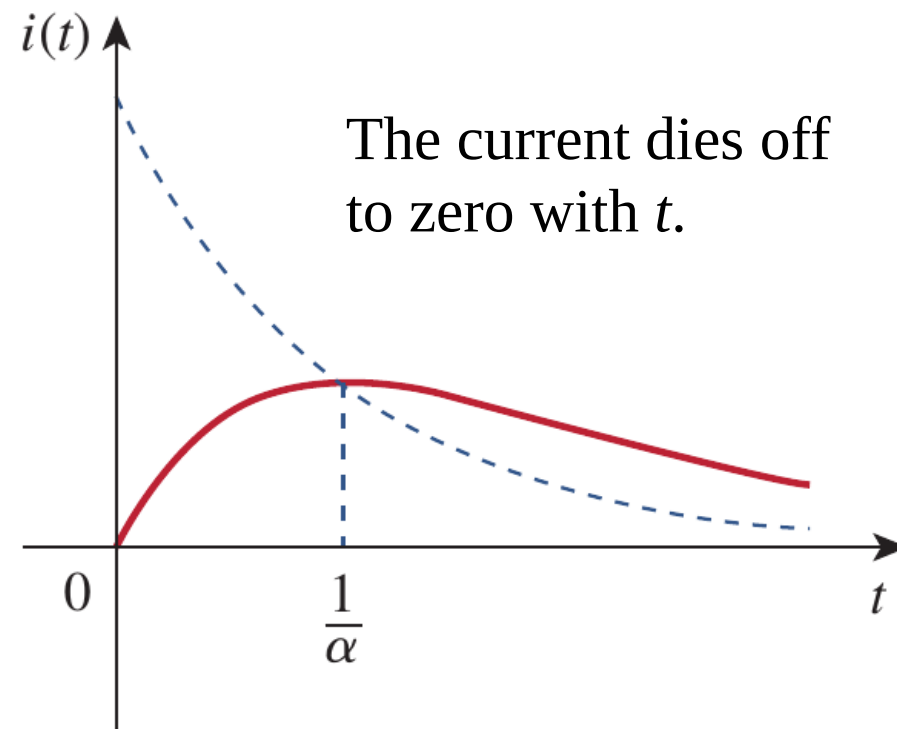
$$\left(\frac{R}{2L}\right)^2 - \frac{1}{LC} = \left(\frac{R}{2L} + \frac{1}{\sqrt{LC}}\right)\left(\frac{R}{2L} - \frac{1}{\sqrt{LC}}\right) = 0$$

b) Therefore  $s_1$  and  $s_2$  are an identical real number with a negative value and the solution is:

$$i(t) = e^{-\alpha t}(A_1 + A_2 t)$$

c) You can use initial values to find  $A_1$  and  $A_2$

$$\alpha = \frac{R}{2L}, \quad \omega_0 = \frac{1}{\sqrt{LC}}$$





## 8.5 Natural response of series RLC circuits (e)

- $\alpha < \omega_0$ : underdamped (欠阻尼)

a) When  $\alpha < \omega_0$ , we have:

$$\left(\frac{R}{2L}\right)^2 - \frac{1}{LC} = \left(\frac{R}{2L} + \frac{1}{\sqrt{LC}}\right)\left(\frac{R}{2L} - \frac{1}{\sqrt{LC}}\right) < 0$$

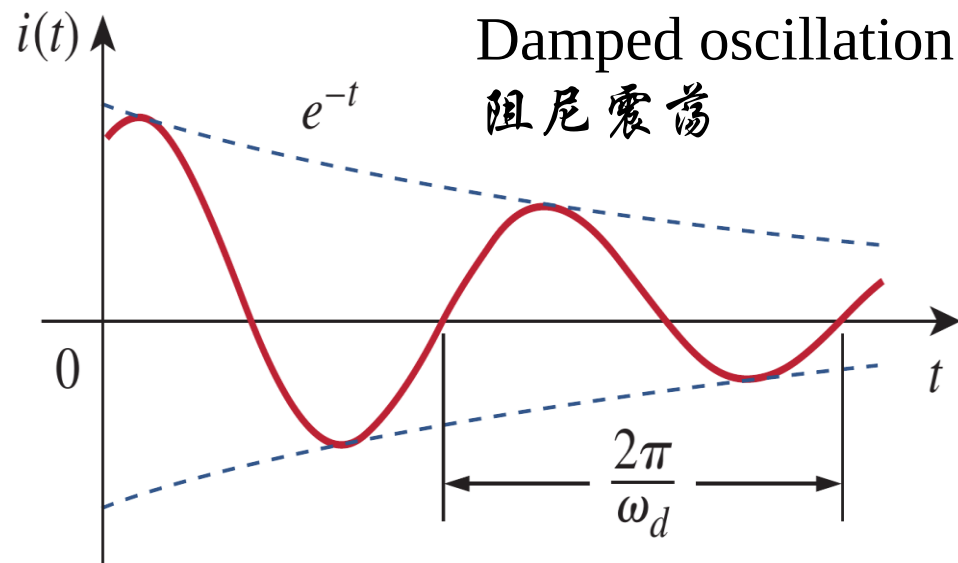
b)  $s_1$  and  $s_2$  are different complex numbers

$$i(t) = e^{-\alpha t} (A_1 \cos \omega_d t + A_2 \sin \omega_d t)$$

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2} \quad \text{damped natural frequency (阻尼频率).}$$

c) You can use initial values to find  $A_1$  and  $A_2$

$$\alpha = \frac{R}{2L}, \quad \omega_0 = \frac{1}{\sqrt{LC}}$$



## 8.5 Natural response of series RLC circuits (f)

- $\alpha = 0$ : undamped (无阻尼震荡)

a) When  $R = 0$  ( $LC$  tank circuits),  $\alpha = 0$

b) When  $\alpha = 0$ , we have:  $\left(\frac{R}{2L}\right)^2 - \frac{1}{LC} = -\frac{1}{LC}$

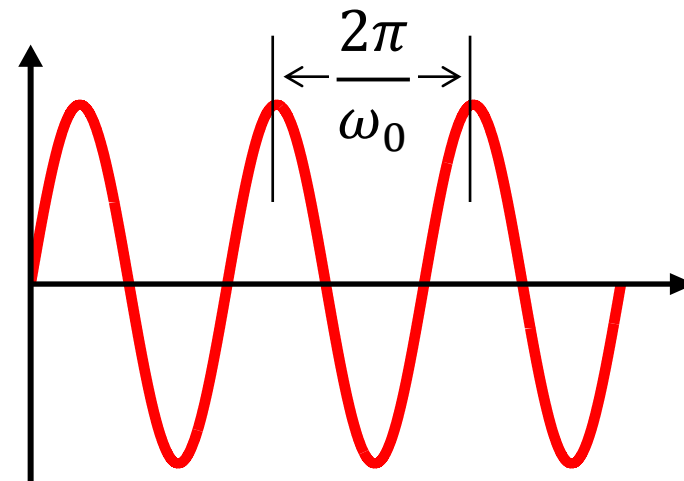
c) Therefore  $s_1 = j\omega_d$  and  $s_2 = -j\omega_d$

$$i(t) = A_1 \cos \omega_0 t + A_2 \sin \omega_0 t$$

d) You can use initial values to find  $A_1$  and  $A_2$

$$\alpha = \frac{R}{2L}, \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

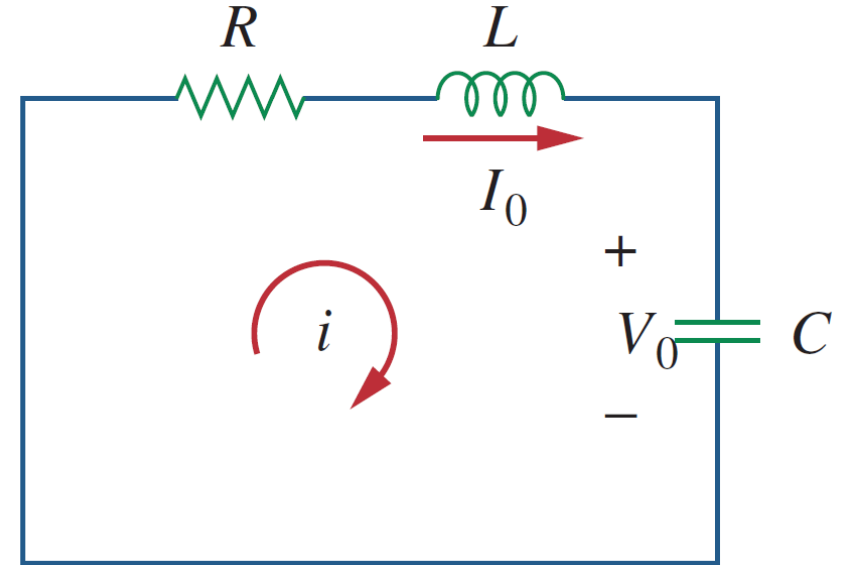
Undamped oscillation  
无阻尼震荡



## 8.5 Natural response of series RLC circuits (g)

- To determine constants  $A_1$  and  $A_2$ :

- a) The constants in each case can be determined from the initial conditions.
- b) We need to know  $i(0)$  and  $i'(0)$ .
- c)  $i(0)$  is the initial current through the inductor.
- d)  $i'(0)$  can be found using KVL.



$$Ri(0) + L \frac{di(0)}{dt} + V_0 = 0 \quad \frac{di(0)}{dt} = -\frac{1}{L}(RI_0 + V_0)$$

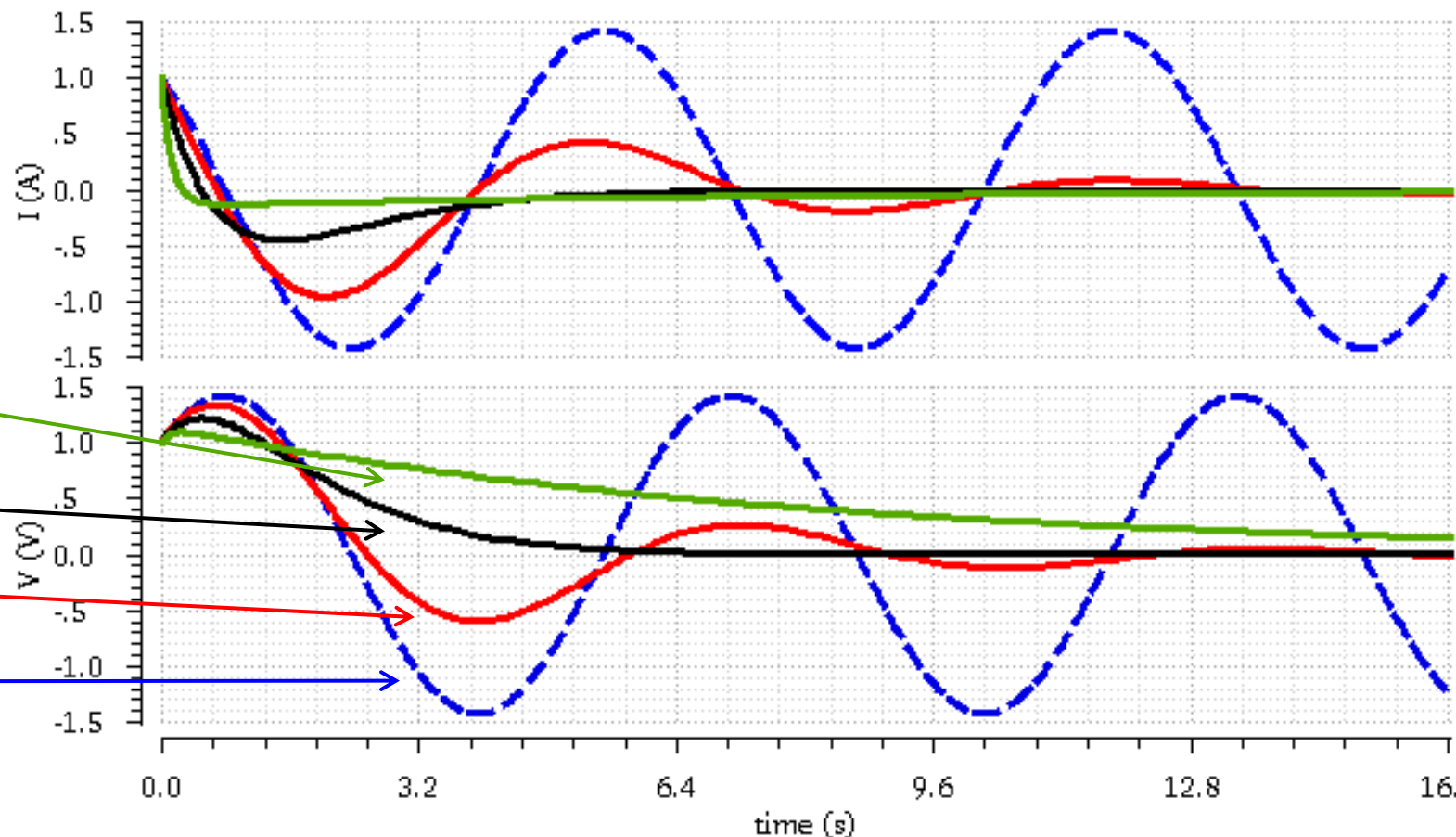
## 8.5 Natural response of series RLC circuits (h)

- Natural response of series RCL circuits:

$$\alpha = \frac{R}{2L}$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

Overdamped  
Critically damped  
Underdamped  
Undamped



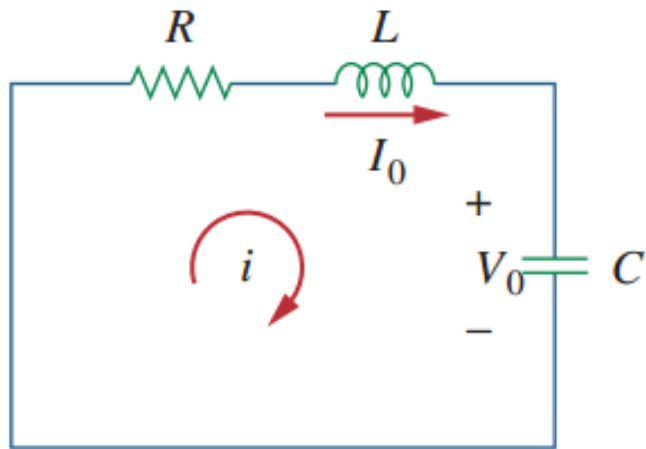
## 8.5 Natural response of series RLC circuits (i)

- **Understanding the behaviors of Series RLC circuit:**
  - **Damping behavior (阻尼):** Representing the gradual loss of the initial stored energy due to the presence of resistor  $R$ . The damping factor  $\alpha$  determines the rate at which the response is damped. By adjusting the value of  $R$ , the response may be made undamped, underdamped, critically damped, or overdamped.
  - **Oscillatory behavior:** Having both type of energy storage elements,  $L$  and  $C$ , in the circuit the response can be oscillatory with loss (underdamped) or without loss (undamped). Having both  $L$  and  $C$  allows for energy transforming between its electric and magnetic forms.
  - **How to achieve undamped oscillation in real life:** Because the real inductor and capacitor are subject to energy loss due to leakage or winding resistance. Undamped response or ideal oscillator can not be implemented just by a simple RLC circuit and needs to use active element such as an op amp.

## 8.5 Natural response of series RLC circuits (j)

- **Example 8.03:**

**Example 8.03:** In the following circuit,  $R = 40\ \Omega$ ,  $L = 4\ \text{H}$ , and  $C = 1/4\ \text{F}$ . Calculate the characteristic roots of the circuit. Is the natural response overdamped, underdamped, or critically damped?



**Solution:**

First calculate the damping factor  $\alpha$  and the resonant frequency  $\omega_0$ . We have

$$\alpha = \frac{R}{2L} = \frac{40}{2 \cdot 4} = 5, \quad \omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{4 \cdot \frac{1}{4}}} = 1$$

The characteristic roots are

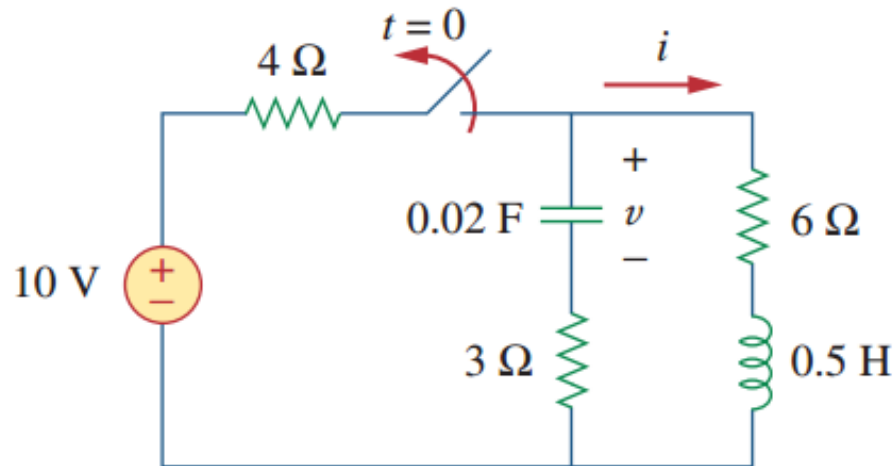
$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} = -5 \pm \sqrt{25 - 1} \\ \rightarrow s_1 = -0.101, s_2 = -9.899$$

Now that  $\alpha > \omega_0$ , the natural response is overdamped.

## 8.5 Natural response of series RLC circuits (k)

- Example 8.04:

**Example 8.04:** Find  $i(t)$  for  $t > 0$  in the circuit on the right. Assume that the circuit has reached steady state at  $t = 0^-$ .



**Solution:**

(1) After switching ( $t > 0$ ), we have a source-free series RLC circuit, in which

$$R = 3 + 6 = 9 \, \Omega, \quad L = 0.5 \, \text{H}, \quad C = 0.02 \, \text{F}.$$

(2) Find the type of natural response:

The damping factor  $\alpha = R/2L = 9 \, \text{rad/s}$

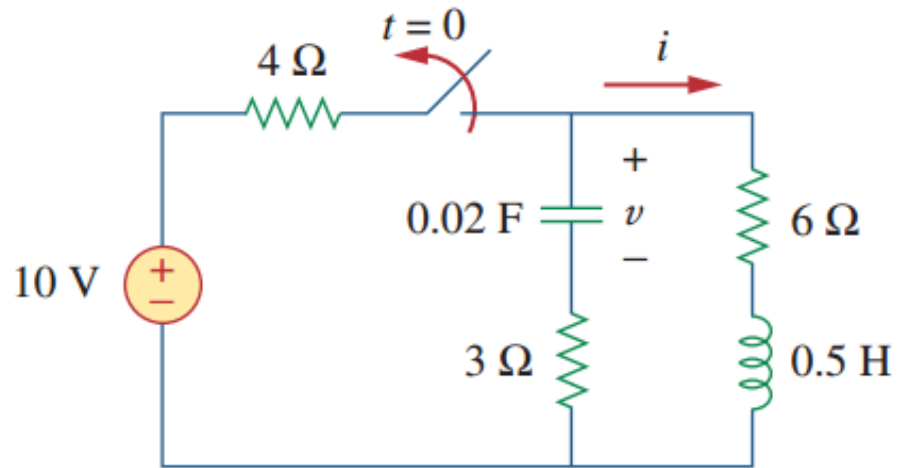
The natural frequency  $\omega_0 = 1/\sqrt{LC} = 10 \, \text{rad/s}$

$\alpha < \omega_0$ , so the circuit is **underdamped**

## 8.5 Natural response of series RLC circuits (I)

- Example 8.04:

**Example 8.04:** Find  $i(t)$  for  $t > 0$  in the circuit on the right. Assume that the circuit has reached steady state at  $t = 0^-$ .



**Solution:**

(3) The you know the solution:

$$\left\{ \begin{array}{l} i(t) = e^{-\alpha t} (A_1 \cos \omega_d t + A_2 \sin \omega_d t) \\ \omega_d = \sqrt{\omega_0^2 - \alpha^2} = 4.359 \\ \alpha = R/2L = 9 \end{array} \right.$$

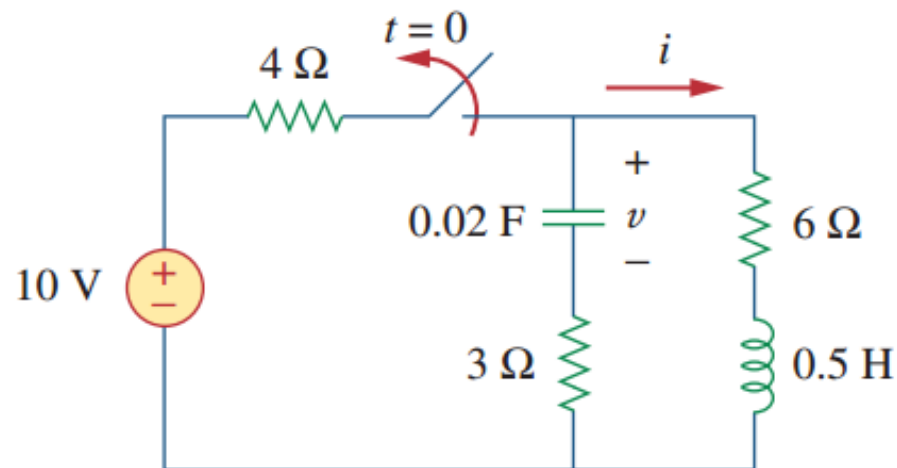
$$i(t) = e^{-9t} (A_1 \cos 4.359t + A_2 \sin 4.359t)$$



## 8.5 Natural response of series RLC circuits (m)

- Example 8.04:

**Example 8.04:** Find  $i(t)$  for  $t > 0$  in the circuit on the right. Assume that the circuit has reached steady state at  $t = 0^-$ .



**Solution:**

$$i(t) = e^{-9t}(A_1 \cos 4.359t + A_2 \sin 4.359t) \quad [1]$$

(4) Use initial values to find  $A_1$  and  $A_2$ :

$$i_L(0^+) = 10/(4 + 6) = 1 \text{ A} \quad [2]$$

$$i'(0^+) = -\frac{Ri(0) + v(0)}{L} = -6 \text{ A/s} \quad [3]$$

Substitute  $t = 0$  and [2] into [1]:  $A_1 = i(0) = 1$

Combine  $t = 0$  and [3] and [1]:

$$-6 = -9(A_1 + 0) + 4.359(-0 + A_2) \Rightarrow A_2 = 0.6882$$

$$(5) \quad i(t) = e^{-9t}(\cos 4.359t + 0.6882 \sin 4.359t) \text{ A}$$

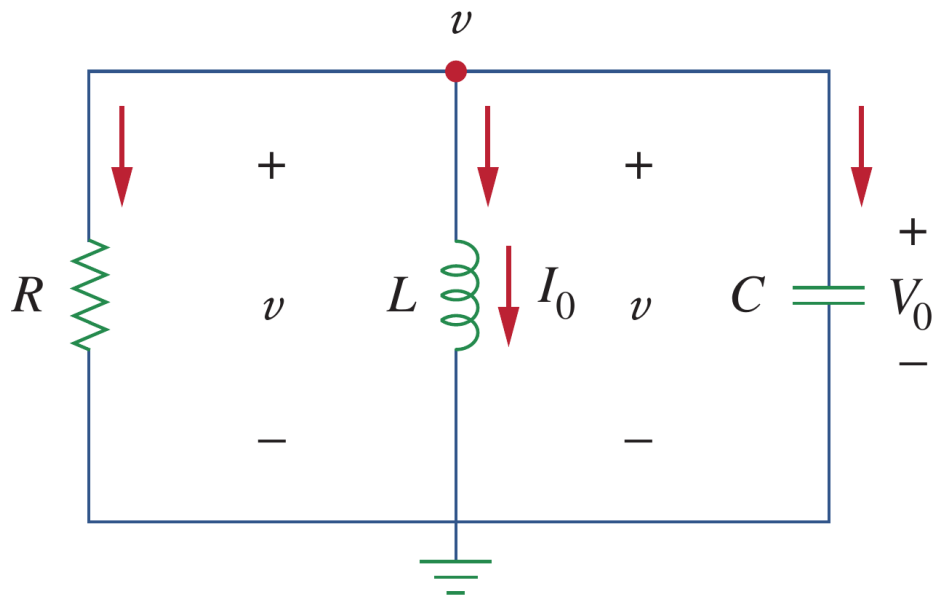
## 8.5 Natural response of series RLC circuits (n)

- Procedures to resolve the natural response:

|        |                         |                                                                                                                                                                                                                                  |
|--------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| STEP 1 | $\alpha$ and $\omega_0$ | Find out the $\alpha$ and the $\omega_0$ .                                                                                                                                                                                       |
| STEP 2 | Damping?                | Is response is over, under, or critically damped?                                                                                                                                                                                |
| STEP 3 | Solution                | Select a solution based on the damping and find $s_1$ and $s_2$ if needed:<br>$v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} \quad v(t) = e^{-\alpha t} (A_1 + A_2 t)$ $v(t) = e^{-\alpha t} (A_1 \cos \omega_d t + A_2 \sin \omega_d t)$ |
| STEP 4 | Initial values          | Find out the initial values, for example $v(0)$ and $v'(0)$ .                                                                                                                                                                    |
| STEP 5 | Coefficients            | Find out $A_1$ and $A_2$ by combining results in steps 3 and 4.                                                                                                                                                                  |

## 8.6 Natural response of parallel RLC circuits (a)

- Natural response of parallel RCL circuits



a) Excited from the initial stored energy: voltage across the capacitor  $V_0$  and/or current through the inductor  $I_0$ .

b) Using KCL:

$$\frac{v}{R} + \frac{1}{L} \int_{-\infty}^t v(\tau) d\tau + C \frac{dv}{dt} = 0$$

$$\frac{d^2v}{dt^2} + \frac{1}{RC} \frac{dv}{dt} + \frac{1}{LC} v = 0$$

Which is a homogenous second-order DE

## 8.6 Natural response of parallel RLC circuits (b)

- The resonant frequency and the damping factor

The homogenous second-order DE:  $\frac{d^2v}{dt^2} + \frac{1}{RC} \frac{dv}{dt} + \frac{1}{LC} v = 0$       The characteristic equation:  $s^2 + \frac{1}{RC} s + \frac{1}{LC} = 0$

$$\Rightarrow s_{1,2} = -\frac{1}{2RC} \pm \sqrt{\left(\frac{1}{2RC}\right)^2 - \frac{1}{LC}} \quad \Rightarrow s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

$$\alpha = \frac{1}{2RC}, \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

- a) The roots  $s_1$  and  $s_2$  are called **natural frequencies**, measured in nepers per second (Np/s).
- b)  $\omega_0$  is the **resonant frequency**, expressed in radians per second (rad/s);
- c)  $\alpha$  is the neper frequency or the **damping factor** (阻尼系数), expressed in nepers per second.

## 8.6 Natural response of parallel RLC circuits (c)

- **Natural response of parallel RLC circuits**

Exactly the same as the case of series RLC, the response can be classified into three categories, depending on their damping or oscillating behaviors.

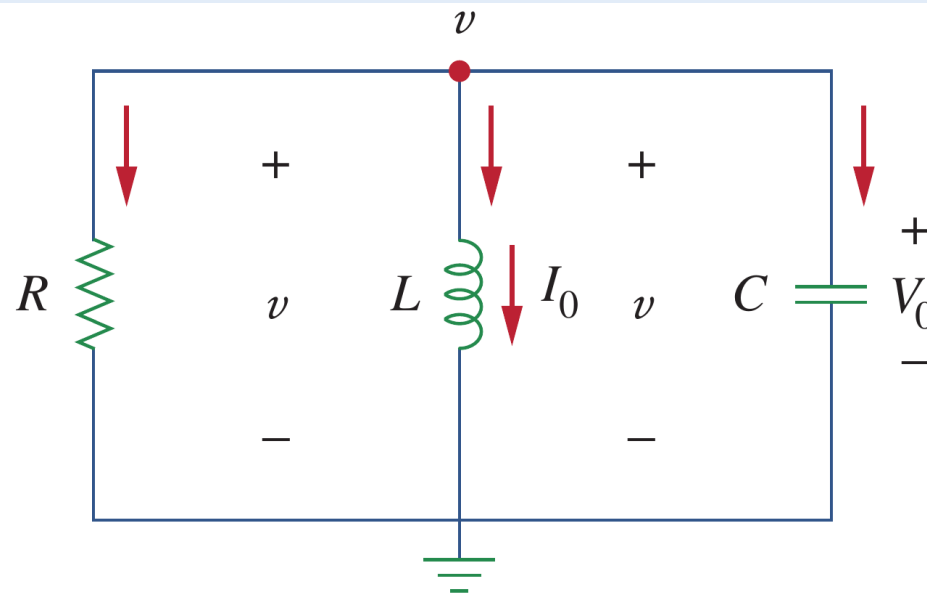
- **overdamped (过阻尼):  $\alpha > \omega_0$** , the two roots  $s_1$  and  $s_2$  are different negative real numbers. The solution  $v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$  dies off to zero with  $t$ .
- **critically damped (临界阻尼)  $\alpha = \omega_0$** , the two roots are identical and have a negative real value,  $s_1 = s_2 = -\alpha$ . The solution  $v(t) = e^{-\alpha t} (A_1 + A_2 t)$ .
- **underdamped (欠阻尼):  $\alpha < \omega_0$** , the two roots are different complex number with negative real part,  $s_1 = -\alpha + j\omega_d$  and  $s_2 = -\alpha - j\omega_d$ . The solution  $v(t) = e^{-\alpha t} (A_1 \cos \omega_d t + A_2 \sin \omega_d t)$ , where  $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$  is called damped natural frequency (阻尼频率).
- **undamped (无阻尼):  $\alpha = 0$** , the two roots are different imaginary numbers,  $s_1 = j\omega_0$  and  $s_2 = -j\omega_0$ . The solution  $v(t) = A_1 \cos \omega_0 t + A_2 \sin \omega_0 t$ .

## 8.6 Natural response of parallel RLC circuits (d)

- To determine constants  $A_1$  and  $A_2$ :

- The constants in each case can be determined from the initial conditions.
- We need to know  $v(0)$  and  $v'(0)$ .
- $v(0)$  is the initial voltage across the capacitor.
- $dv(0)/dt$  can be found using KCL.

$$v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} \quad v(t) = e^{-\alpha t} (A_1 + A_2 t)$$
$$v(t) = e^{-\alpha t} (A_1 \cos \omega_d t + A_2 \sin \omega_d t)$$



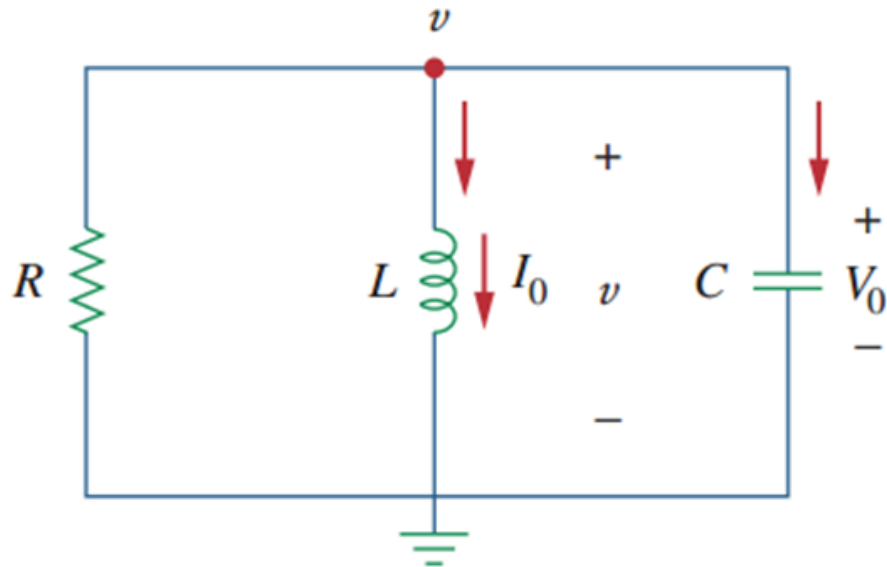
$$\frac{V_0}{R} + I_0 + C \frac{dv(0)}{dt} = 0$$

$$\frac{dv(0)}{dt} = -\frac{(V_0 + RI_0)}{RC}$$

## 8.6 Natural response of parallel RLC circuits (e)

- Example 8.05:

**Example 8.05:** In the parallel circuit below, find  $v(t)$  for  $t > 0$ , assuming  $v(0) = 5 \text{ V}$ ,  $i(0) = 0$ ,  $L = 1 \text{ H}$ , and  $C = 10 \text{ mF}$ . Consider these cases:  $R = 1.923 \text{ } \Omega$ ,  $R = 5 \text{ } \Omega$ , and  $R = 6.25 \text{ } \Omega$ .



### Solution:

(1) Find the initial values and derivatives

The initial voltage is:  $v(0) = 5 \text{ V}$

The initial derivative of voltage is:

$$v'(0) = -[v(0) + i(0)R] / RC$$

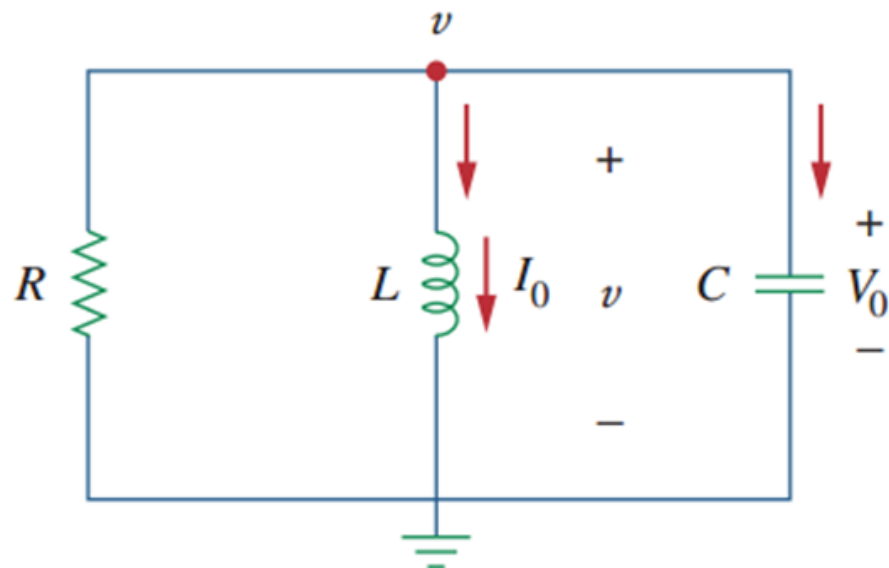
For  $R = 1.923 \Omega$ ,  $5 \Omega$ , and  $6.25 \Omega$ :

| Case | $R$   | $v(0)$ | $v'(0)$ |
|------|-------|--------|---------|
| 1    | 1.923 | 5      | -260    |
| 2    | 5     | 5      | -100    |
| 3    | 6.25  | 5      | -80     |

## 8.6 Natural response of parallel RLC circuits (f)

- Example 8.05:

**Example 8.05:** In the parallel circuit below, find  $v(t)$  for  $t > 0$ , assuming  $v(0) = 5$  V,  $i(0) = 0$ ,  $L = 1$  H, and  $C = 10$  mF. Consider these cases:  $R = 1.923$   $\Omega$ ,  $R = 5$   $\Omega$ , and  $R = 6.25$   $\Omega$ .



**Solution:**

(2) Find how the circuit is damped

The resonant frequency  $\omega_0 = 1/\sqrt{LC} = 10$  rad/s

The damping factor  $\alpha = 1/2RC$

| Case | $R$   | $\alpha$ | $\omega_0$ |                   |
|------|-------|----------|------------|-------------------|
| 1    | 1.923 | 26       | 10         | Overdamped        |
| 2    | 5     | 10       | 10         | Critically damped |
| 3    | 6.25  | 8        | 10         | Underdamped       |



## 8.6 Natural response of parallel RLC circuits (g)

- Example 8.05:

Solution:

| Case | $R$   | $v(0)$ | $v'(0)$ | $\alpha$ | $\omega_0$ | Natural response  | Solution                                                           |
|------|-------|--------|---------|----------|------------|-------------------|--------------------------------------------------------------------|
| 1    | 1.923 | 5      | -260    | 26       | 10         | Overdamped        | $v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$                             |
| 2    | 5     | 5      | -100    | 10       | 10         | Critically damped | $v(t) = e^{-\alpha t} (A_1 + A_2 t)$                               |
| 3    | 6.25  | 5      | -80     | 8        | 10         | Underdamped       | $v(t) = e^{-\alpha t} (A_1 \cos \omega_d t + A_2 \sin \omega_d t)$ |

Now all you need to do is put these values into the corresponding solutions, and then use the initial values to calculate  $A_1$  and  $A_2$  in each case

## 8.6 Natural response of parallel RLC circuits (h)

- Example 8.05:

Solution: (3) Case one  $R = 1.923 \Omega$

The damping factor  $\alpha = 1/2RC = 1/(2 \times 1.923 \times 10e^{-3}) = 26 > \omega_0$ . This is an **overdamped case**, and the response for  $t > 0$

$$v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} \quad \text{where } s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} = -2; \quad s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2} = -50$$

Use initial values to find  $A_1$  and  $A_2$ :

$$\begin{cases} v(0) = 5 \text{ V} \\ v'(0) = -260 \text{ V/s} \end{cases} \rightarrow \begin{cases} 5 = (A_1 \cdot 1 + A_2 \cdot 1) \\ -260 = A_1 \cdot s_1 \cdot 1 + A_2 \cdot s_2 \cdot 1 = -2A_1 - 50A_2 \end{cases} \rightarrow \begin{cases} A_1 = -0.2083 \\ A_2 = 5.2083 \end{cases}$$

Therefore the voltage across the capacitor is:  $v(t) = -0.2083e^{-2t} + 5.2083e^{-50t}$

## 8.6 Natural response of parallel RLC circuits (i)

- Example 8.05:

Solution: (4) Case two  $R = 5 \Omega$

$\alpha = \frac{1}{2RC} = \frac{1}{2 \times 5 \times 10 \times 10^{-3}} = 10 = \omega_0$ , so the natural response is **critically damped** and

$$v(t) = e^{-\alpha t} (A_1 + A_2 t)$$

Use initial values to find  $A_1$  and  $A_2$ :

$$\begin{cases} v(0) = 5 \text{ V} \\ v'(0) = -100 \text{ V/s} \end{cases} \Rightarrow \begin{cases} 5 = 1 \cdot (A_1 \cdot 1 + A_2 \cdot 0) \\ -100 = -10 \cdot (A_1 \cdot 1 + A_2 \cdot 0) + 1 \cdot (0 + A_2) = -10A_1 + A_2 \end{cases} \Rightarrow \begin{cases} A_1 = 5 \\ A_2 = -50 \end{cases}$$

Therefore the voltage across the capacitor is:  $v(t) = e^{-10t} (5 - 50t)$

## 8.6 Natural response of parallel RLC circuits (j)

- Example 8.05:

Solution: (5) Case three  $R = 6.25 \Omega$

$$\alpha = \frac{1}{2RC} = \frac{1}{2 \times 6.25 \times 10 \times 10^{-3}} = 8 < \omega_0, \text{ so the natural response is underdamped and}$$

$$v(t) = e^{-\alpha t} (A_1 \cos \omega_d t + A_2 \sin \omega_d t), \text{ where } \omega_d = \sqrt{\omega_0^2 - \alpha^2} = 6$$

Use initial values to find  $A_1$  and  $A_2$ :

$$\begin{cases} v(0) = 5 \text{ V} \\ v'(0) = -80 \text{ V/s} \end{cases} \rightarrow \begin{cases} 5 = 1 \cdot (A_1 \cdot 1 + A_2 \cdot 0) \\ -80 = 1 \cdot (-6A_1 \cdot 0 + 6A_2 \cdot 1 - 8A_1 \cdot 1 - 8A_2 \cdot 0) = -8A_1 + 6A_2 \end{cases} \rightarrow \begin{cases} A_1 = 5 \\ A_2 = -6.667 \end{cases}$$

Therefore the voltage across the capacitor is:  $v(t) = e^{-8t} (5 \cos 6t - 6.667 \sin 6t)$

## 8.6 Natural response of parallel RLC circuits (k)

- Procedures to resolve the natural response, example 8.05:

|       | STEP 1   |            | STEP 2            | STEP 3                                                             | STEP 4 |         | STEP 5  |        |
|-------|----------|------------|-------------------|--------------------------------------------------------------------|--------|---------|---------|--------|
| $R$   | $\alpha$ | $\omega_0$ | Natural response  | Solution                                                           | $v(0)$ | $v'(0)$ | $A_1$   | $A_2$  |
| 1.923 | 26       | 10         | Overdamped        | $v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$                             | 5      | -260    | -0.2083 | 5.2083 |
| 5     | 10       | 10         | Critically damped | $v(t) = e^{-\alpha t} (A_1 + A_2 t)$                               | 5      | -100    | 5       | -50    |
| 6.25  | 8        | 10         | Underdamped       | $v(t) = e^{-\alpha t} (A_1 \cos \omega_d t + A_2 \sin \omega_d t)$ | 5      | -80     | 5       | -6.667 |

# Lesson 8 Second-order circuits

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|         |                       |              |
|---------|-----------------------|--------------|
| 8.1-8.3 | Second-order circuits | Slides 08-20 |
|---------|-----------------------|--------------|

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|         |                                  |              |
|---------|----------------------------------|--------------|
| 8.4-8.6 | Natural response of RLC circuits | Slides 22-50 |
|---------|----------------------------------|--------------|

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|                |                                      |                     |
|----------------|--------------------------------------|---------------------|
| <b>8.7-8.8</b> | <b>Step response of RLC circuits</b> | <b>Slides 52-70</b> |
|----------------|--------------------------------------|---------------------|

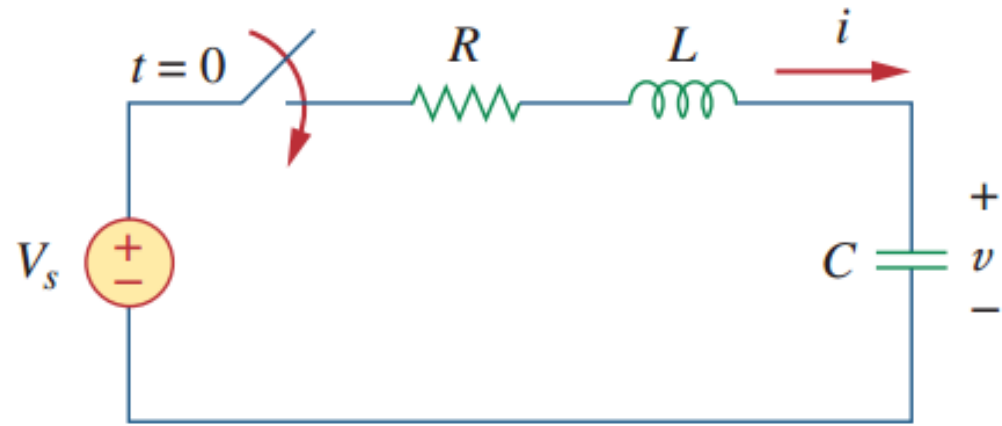
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|     |         |              |
|-----|---------|--------------|
| 8.9 | Summary | Slides 72-73 |
|-----|---------|--------------|

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## 8.7 Step response of series RLC circuits (a)

- Series RLC circuits



$$\begin{cases} L \frac{di}{dt} + Ri + v = V_s \\ i = C \frac{dv}{dt} \end{cases}$$



$$\frac{d^2v}{dt^2} + \frac{R}{L} \frac{dv}{dt} + \frac{v}{LC} = \frac{V_s}{LC}$$

- Consider the series RLC circuit shown on the right. The voltage source is switched on at  $t = 0$ . The initial stored energy at  $t = 0$  are specified by  $V_0$  and  $I_0$ .
- Applying KVL, for  $t > 0$  we have

## 8.7 Step response of series RLC circuits (b)

- Series RLC circuits

$$\frac{d^2v}{dt^2} + \frac{R}{L} \frac{dv}{dt} + \frac{v}{LC} = \frac{V_s}{LC}$$

- This is a nonhomogeneous (非齐次) second-order DE, whose general solution is of the same form for the source-free series RLC circuit.
- The RHS is of polynomial type, particular solution is simply  $v(t) = v_s$ . The total solution is then given by:

$$v(t) = V_s + A_1 e^{s_1 t} + A_2 e^{s_2 t} \quad (\text{Overdamped})$$

$$v(t) = V_s + (A_1 + A_2 t) e^{-\alpha t} \quad (\text{Critically damped})$$

$$v(t) = V_s + (A_1 \cos \omega_d t + A_2 \sin \omega_d t) e^{-\alpha t} \quad (\text{Underdamped})$$



## 8.7 Step response of series RLC circuits (c)

- Total step response of series RLC circuits

**Overdamped:**

$$v(t) = V_s + A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

**Critically damped:**

$$v(t) = V_s + (A_1 + A_2 t) e^{-\alpha t}$$

**Underdamped:**

$$v(t) = V_s + (A_1 \cos \omega_d t + A_2 \sin \omega_d t) e^{-\alpha t}$$

$$v(t) = v_{ss}(t) + v_t(t)$$



**Complete  
response**

=

**Steady-state response**

Permanent part  $v(\infty)$   
After all transients are done

+

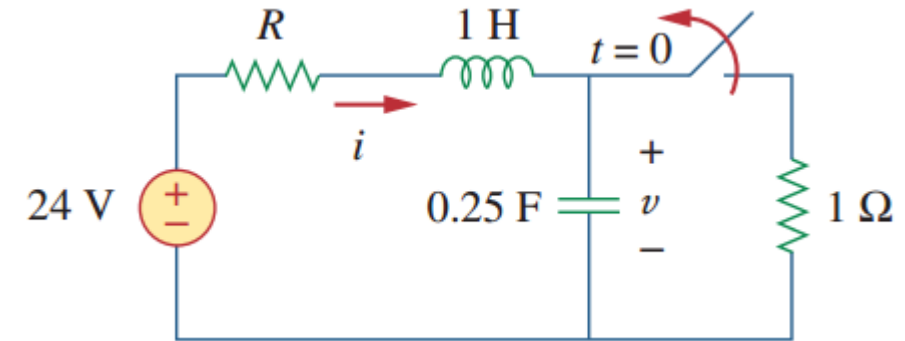
**Transient response**

Temporary part  
Decaying to zero

## 8.7 Step response of series RLC circuits (d)

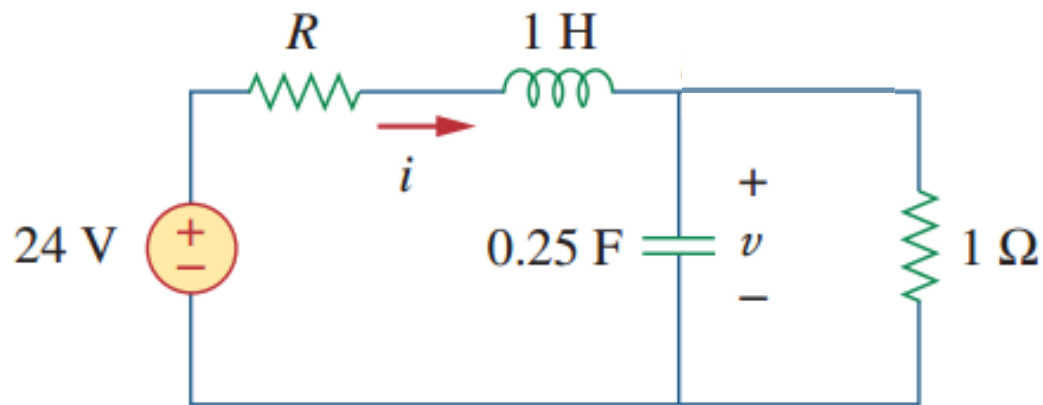
### • Example 8.06

**Example 8.06:** For the following circuit, find  $v(t)$  and  $i(t)$  for  $t > 0$ . Consider these cases:  $R = 5 \Omega$ ,  $R = 4 \Omega$ , and  $R = 1 \Omega$ .



**Solution:** (1) Find initial values

The initial equivalent circuit:



The initial voltage  $v(0) = 24 \times 1/(1 + R)$ .

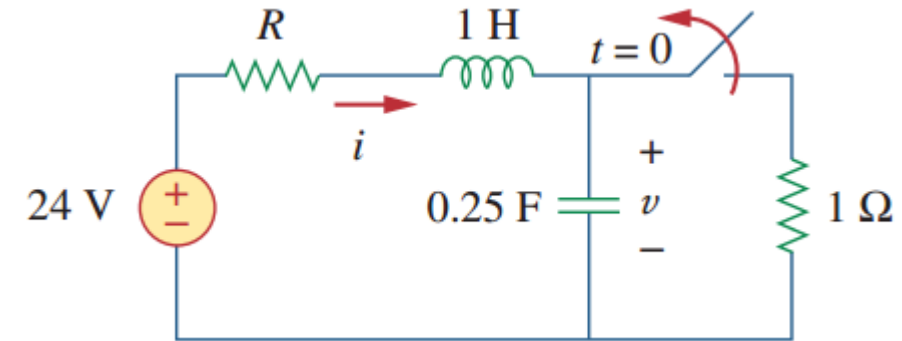
The initial current  $i(0) = 24/(1 + R)$ .

| Case | $R$ | $v(0)$ | $i(0)$ |
|------|-----|--------|--------|
| 1    | 5   | 4      | 4      |
| 2    | 4   | 4.8    | 4.8    |
| 3    | 1   | 12     | 12     |

## 8.7 Step response of series RLC circuits (e)

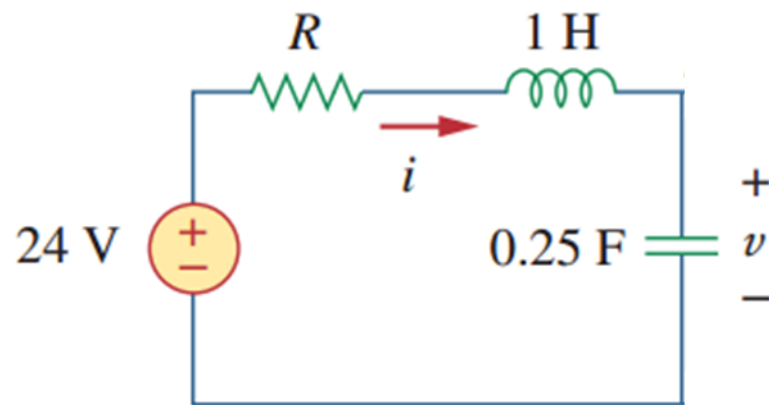
### • Example 8.06

**Example 8.06:** For the following circuit, find  $v(t)$  and  $i(t)$  for  $t > 0$ . Consider these cases:  $R = 5 \Omega$ ,  $R = 4 \Omega$ , and  $R = 1 \Omega$ .



**Solution:** (2) Find initial derivatives and final values

The final equivalent circuit:



At  $t = 0$  when the switch is open, the current through the inductor can not change:

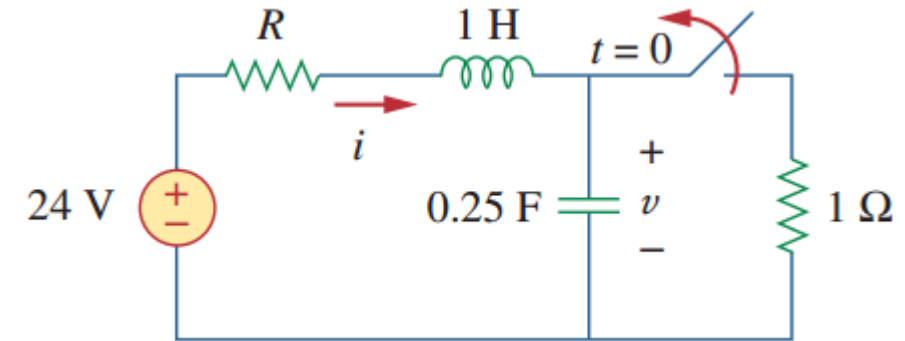
$$i(0) = \frac{24}{1 + R} = C \frac{dv(0)}{dt} \Rightarrow \frac{dv(0)}{dt} = \frac{24}{0.25(1 + R)}$$

| Case | $R$ | $v(0)$ | $v'(0)$ | $v(\infty)$ |
|------|-----|--------|---------|-------------|
| 1    | 5   | 4      | 16      | 24          |
| 2    | 4   | 4.8    | 19.2    | 24          |
| 3    | 1   | 12     | 48      | 24          |

## 8.7 Step response of series RLC circuits (f)

- Example 8.06

**Example 8.06:** For the following circuit, find  $v(t)$  and  $i(t)$  for  $t > 0$ . Consider these cases:  $R = 5 \Omega$ ,  $R = 4 \Omega$ , and  $R = 1 \Omega$ .



**Solution:** (3) Find how the circuit is damped

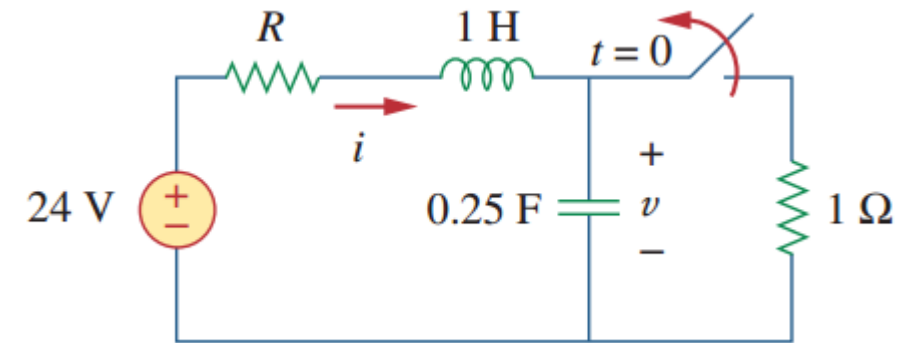
The resonant frequency  $\omega_0 = 1/\sqrt{LC} = 2 \text{ rad/s}$ , The damping factor  $\alpha = R/2L$

| Case | $R$ | $\alpha$ | $\omega_0$ |                   |
|------|-----|----------|------------|-------------------|
| 1    | 5   | 2.5      | 2          | Overdamped        |
| 2    | 4   | 2        | 2          | Critically damped |
| 3    | 1   | 0.5      | 2          | Underdamped       |

## 8.7 Step response of series RLC circuits (g)

- Example 8.06

**Example 8.06:** For the following circuit, find  $v(t)$  and  $i(t)$  for  $t > 0$ . Consider these cases:  $R = 5 \Omega$ ,  $R = 4 \Omega$ , and  $R = 1 \Omega$ .



**Solution:**

| Case | $v(\infty)$ | $v(0)$ | $v'(0)$ | $R$ | $\alpha$ | $\omega_0$ | Damping  | Solution                                                                 |
|------|-------------|--------|---------|-----|----------|------------|----------|--------------------------------------------------------------------------|
| 1    | 24          | 4      | 16      | 5   | 2.5      | 2          | over     | $v(t) = V_s + A_1 e^{s_1 t} + A_2 e^{s_2 t}$                             |
| 2    | 24          | 4.8    | 19.2    | 4   | 2        | 2          | critical | $v(t) = V_s + (A_1 + A_2 t) e^{-\alpha t}$                               |
| 3    | 24          | 12     | 48      | 1   | 0.5      | 2          | under    | $v(t) = V_s + (A_1 \cos \omega_d t + A_2 \sin \omega_d t) e^{-\alpha t}$ |

For each case, you need to put these values into the solution, and then use the initial values to calculate  $A_1$  and  $A_2$  in each case

## 8.7 Step response of series RLC circuits (h)

- Example 8.06

Solution: (4) Case one  $R = 5 \Omega$

| Case | $R$ | $\alpha$ | $\omega_0$ | $v(\infty)$ | $v(0)$ | $v'(0)$ | Damping | Solution                                     |
|------|-----|----------|------------|-------------|--------|---------|---------|----------------------------------------------|
| 1    | 5   | 2.5      | 2          | 24          | 4      | 16      | over    | $v(t) = V_s + A_1 e^{s_1 t} + A_2 e^{s_2 t}$ |

$$v(t) = 24 + A_1 e^{s_1 t} + A_2 e^{s_2 t}, \text{ and } s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} = -1; s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2} = -4$$

Substituting the initial conditions, we get:

$$\begin{cases} 4 = (24 + A_1 \cdot 1 + A_2 \cdot 1) \\ 16 = A_1 \cdot (-1) \cdot 1 + A_2 \cdot (-4) \cdot 1 \end{cases} \rightarrow A_1 = -\frac{64}{3}, A_2 = \frac{4}{3}$$

Finally:  $v(t) = 24 - \frac{64}{3} e^{-t} + \frac{4}{3} e^{-4t} \Rightarrow i(t) = C \frac{dv}{dt} \Rightarrow i(t) = \frac{4}{3} (4e^{-t} - e^{-4t}) \text{ A}$

## 8.7 Step response of series RLC circuits (i)

- Example 8.06

Solution: (5) Case two  $R = 4 \Omega$

| Case | $R$ | $\alpha$ | $\omega_0$ | $v(\infty)$ | $v(0)$ | $v'(0)$ | Damping  | Solution                                  |
|------|-----|----------|------------|-------------|--------|---------|----------|-------------------------------------------|
| 2    | 4   | 2        | 2          | 24          | 4.8    | 19.2    | critical | $v(t) = V_s + (A_1 + A_2 t)e^{-\alpha t}$ |

$$v(t) = 24 + e^{-\alpha t}(A_1 + A_2 t)$$

Substituting the initial conditions, we get:

$$\begin{cases} 4.8 = 24 + 1 \cdot (A_1 + A_2 \cdot 0) \\ 19.2 = -2 \cdot (A_1 + A_2 \cdot 0) + 1 \cdot A_2 \end{cases} \rightarrow A_1 = -19.2, A_2 = -19.2$$

Finally:  $v(t) = 24 - 19.2e^{-2t}(1+t) \Rightarrow i(t) = C \frac{dv}{dt} \Rightarrow i(t) = (4.8 + 9.6t)e^{-2t} \text{ A}$

## 8.7 Step response of series RLC circuits (j)

- Example 8.06

Solution: (6) Case three  $R = 1 \Omega$

| Case | $R$ | $\alpha$ | $\omega_0$ | $v(\infty)$ | $v(0)$ | $v'(0)$ | Damping | Solution                                                                |
|------|-----|----------|------------|-------------|--------|---------|---------|-------------------------------------------------------------------------|
| 3    | 1   | 0.5      | 2          | 24          | 12     | 48      | under   | $v(t) = V_s + (A_1 \cos \omega_d t + A_2 \sin \omega_d t)e^{-\alpha t}$ |

$$v(t) = 24 + e^{-0.5t}(A_1 \cos \omega_d t + A_2 \sin \omega_d t) \quad \omega_d = \sqrt{\omega_0^2 - \alpha^2} = 1.936$$

Substituting the initial conditions, we get: 
$$\begin{cases} 12 = 24 + 1 \cdot (A_1 \cdot 1 + A_2 \cdot 0) \\ 48 = -0.5 \cdot (A_1 \cdot 1 + A_2 \cdot 0) + 1 \cdot (0 + 1.936 A_2 \cdot 1) \end{cases} \rightarrow A_1 = -12, A_2 = 21.694$$

Finally: 
$$v(t) = 24 + e^{-0.5t}(-12 \cos 1.936t + 21.694 \sin 1.936t)$$

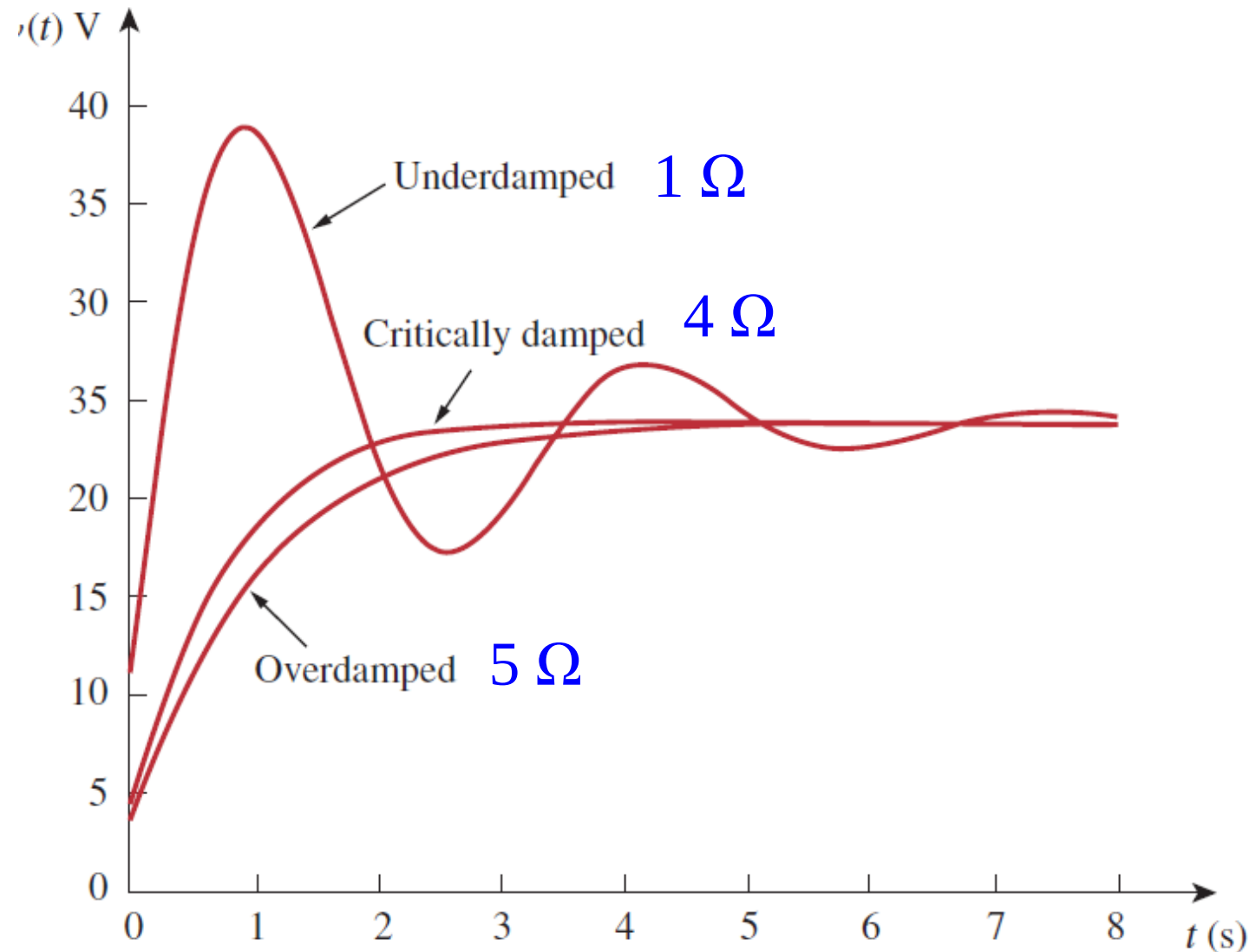
$$i(t) = (3.1 \sin 1.936t + 12 \cos 1.936t)e^{-0.5t} \text{ A}$$



## 8.7 Step response of series RLC circuits (k)

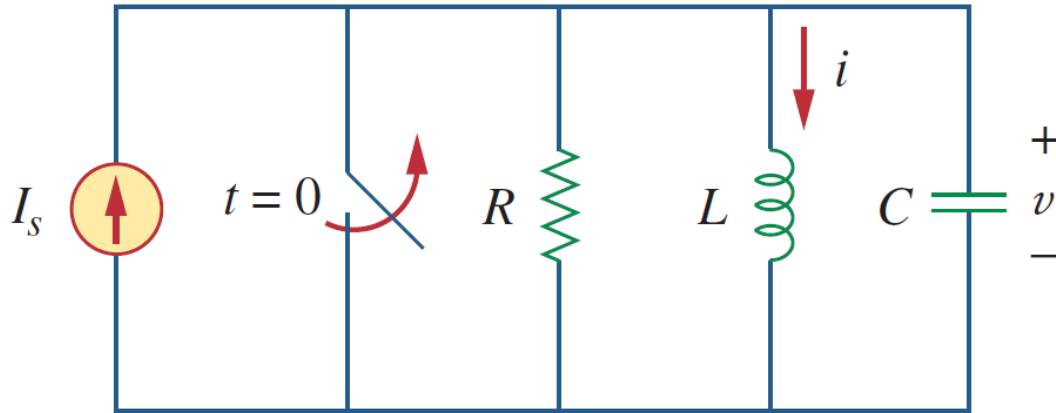
- Example 8.06

Waveforms:



## 8.8 Step response of parallel RLC circuits (a)

- Parallel RLC circuits



$$\begin{cases} \frac{v}{R} + i + C \frac{dv}{dt} = I_s \\ v = L \frac{di}{dt} \end{cases}$$



$$\frac{d^2 i}{dt^2} + \frac{1}{RC} \frac{di}{dt} + \frac{i}{LC} = \frac{I_s}{LC}$$

- a) Consider the parallel RLC circuit shown above. We want to find the response due to a sudden application of a dc current. The initial stored energy at  $t = 0$  are specified by  $V_0$  and  $I_0$ .
- b) Applying KCL, for  $t > 0$  we have

## 8.8 Step response of parallel RLC circuits (b)

- Series RLC circuits

$$\frac{d^2i}{dt^2} + \frac{1}{RC} \frac{di}{dt} + \frac{i}{LC} = \frac{I_s}{LC}$$

- a) This is a nonhomogeneous (非齐次) second-order DE, whose general solution is of the same form for the source-free parallel RLC circuit.
- b) The RHS is of polynomial type, particular solution is simply  $v(t) = v_s$ . The total solution is then given by:

$$i(t) = I_s + A_1 e^{s_1 t} + A_2 e^{s_2 t} \quad (\text{Overdamped})$$

$$i(t) = I_s + (A_1 + A_2 t) e^{-\alpha t} \quad (\text{Critically damped})$$

$$i(t) = I_s + (A_1 \cos \omega_d t + A_2 \sin \omega_d t) e^{-\alpha t} \quad (\text{Underdamped})$$

## 8.8 Step response of parallel RLC circuits (c)

- Total step response of parallel RLC circuits

**Overdamped:**

$$i(t) = I_s + A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

**Critically damped:**

$$i(t) = I_s + (A_1 + A_2 t) e^{-\alpha t}$$

**Underdamped:**

$$i(t) = I_s + (A_1 \cos \omega_d t + A_2 \sin \omega_d t) e^{-\alpha t}$$

$$i(t) = i_{ss}(t) + i_t(t)$$



**Complete  
response**

=

**Steady-state response**

Permanent part  $i(\infty)$   
After all transients are done

+

**Transient response**

Temporary part  
Decaying to zero

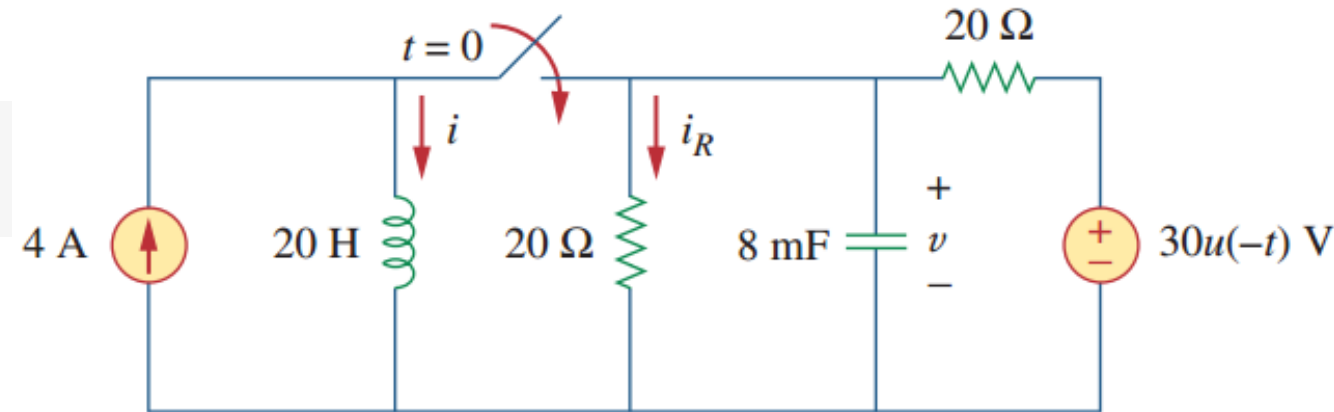
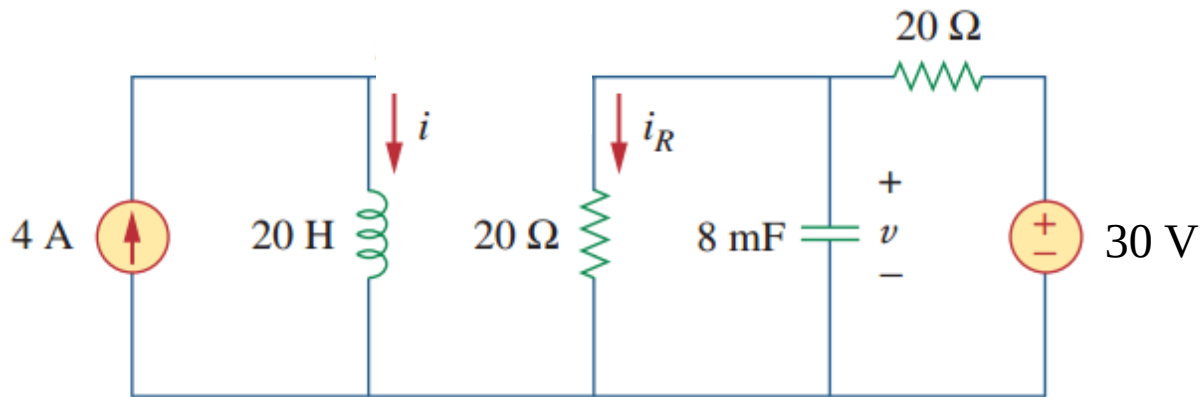
## 8.8 Step response of parallel RLC circuits (d)

- Example 8.07

Example 8.07: find  $i(t)$  and  $i_R(t)$  for  $t > 0$ .

Solution: (1) Analyze the initial circuit and find initial values.

The equivalent circuit for  $t < 0$ :



For  $t < 0$ , the 4-A current goes through the inductor which is short for dc.

$$\text{So } i(0) = 4 \text{ A.}$$

The  $30u(-t)$  is 30 V for  $t < 0$ ,

$$\text{so } v(0) = 30 \cdot 20 / (20 + 20) = 15 \text{ V}$$

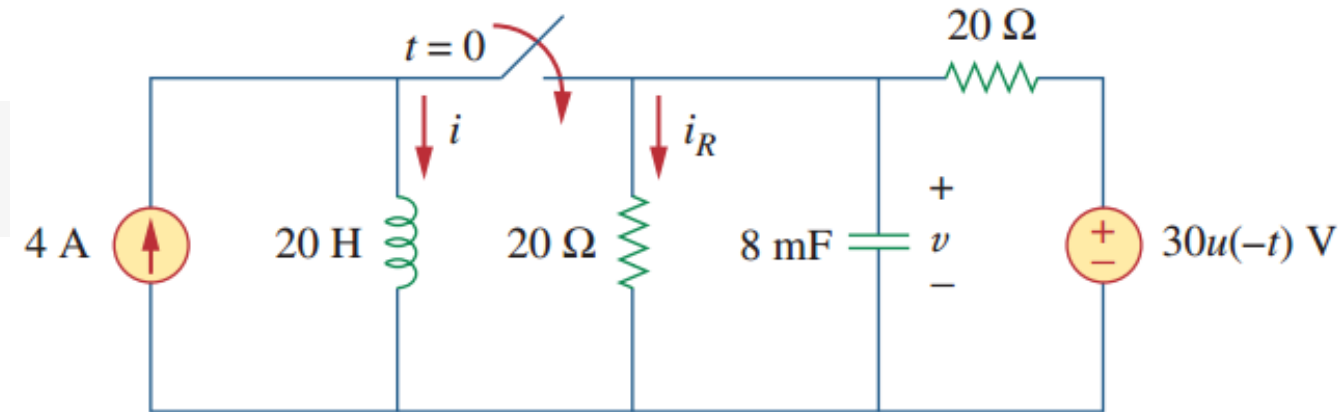
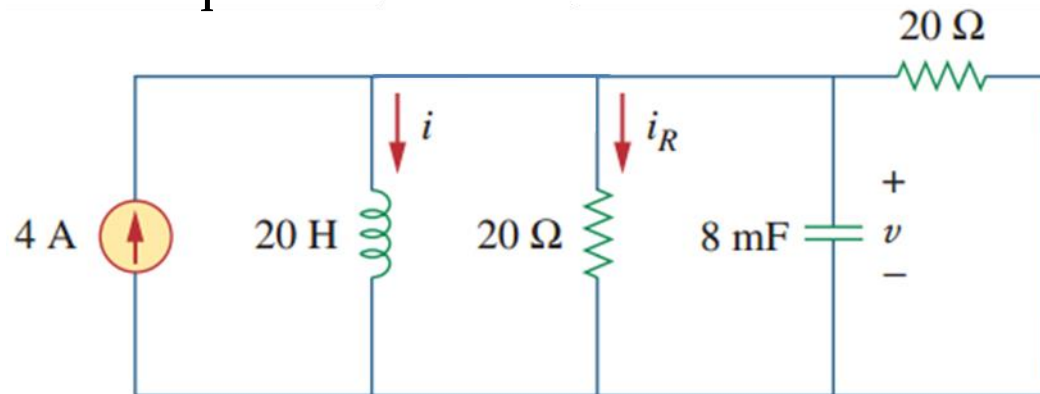
## 8.8 Step response of parallel RLC circuits (e)

- Example 8.07

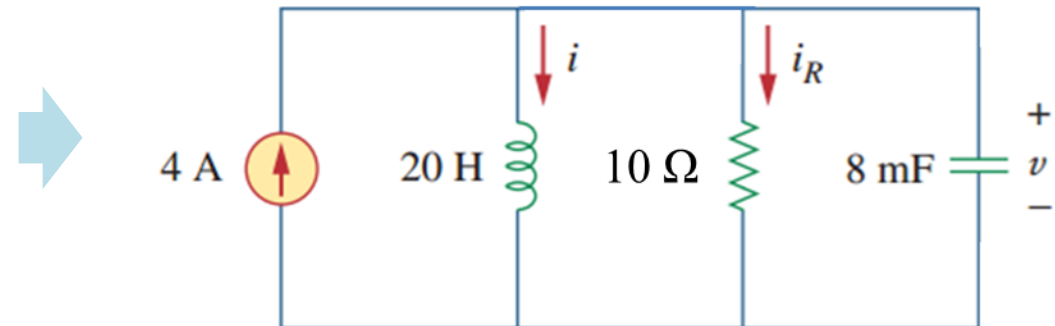
Example 8.07: find  $i(t)$  and  $i_R(t)$  for  $t > 0$ .

Solution: (2) Analyze the final circuit and find initial derivatives and final values.

The equivalent circuit for  $t > 0$ :



$$\begin{cases} v(0) = L \frac{di(0)}{dt} = 15 \text{ V} \Rightarrow i'(0) = 0.75 \text{ A/s} \\ i(\infty) = 4 \text{ A} \end{cases}$$



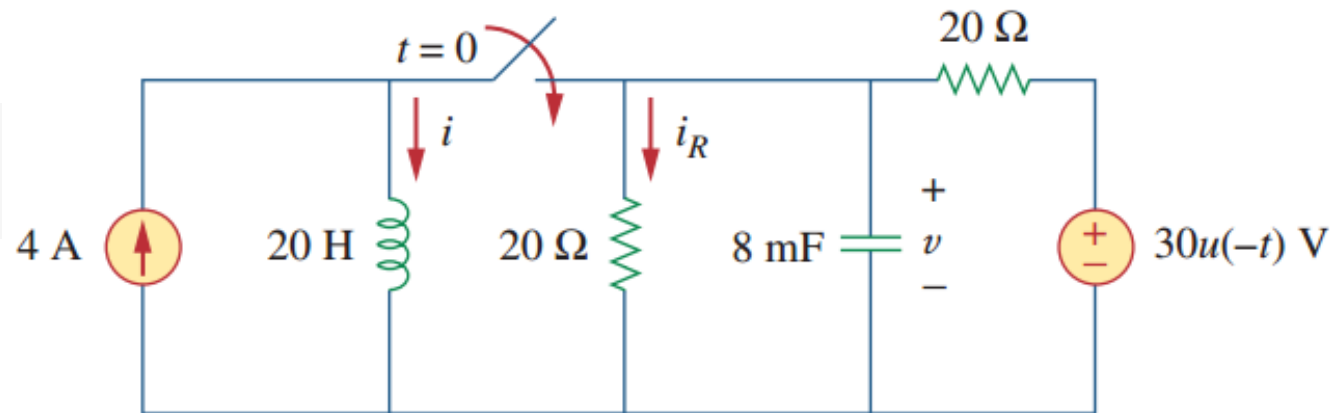
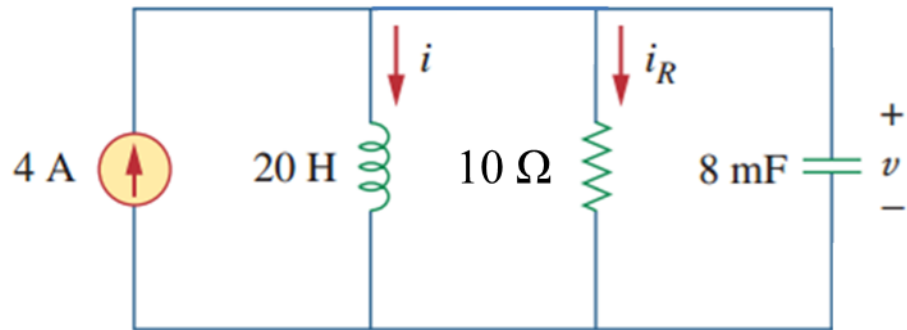
## 8.8 Step response of parallel RLC circuits (f)

- Example 8.07

Example 8.07: find  $i(t)$  and  $i_R(t)$  for  $t > 0$ .

Solution: (3) Based on the final circuit, find  $\alpha$ ,  $\omega_0$ , and how the circuit is damped.

The equivalent circuit for  $t > 0$ :



$$\begin{cases} \alpha = \frac{1}{2RC} = \frac{1}{2 \times 10 \times 8 \times 10^{-3}} = 6.25 \\ \omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{20 \times 8 \times 10^{-3}}} = 2.5, \end{cases}$$

$\alpha > \omega_0$ , so the circuit is over damped.

## 8.8 Step response of parallel RLC circuits (g)

- Example 8.07

Solution: (4) Recall the solution of  $i$  for overdamped parallel RLC circuits, find  $A_1$  and  $A_2$ .

$$i(t) = 4 + A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

where  $s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} = -0.5218$ ,  $s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2} = -11.978$

Using the initial conditions 
$$\begin{cases} 4 = 4 + (A_1 + A_2) \\ 0.75 = (-0.5218A_1 \cdot 1 - 11.978A_2 \cdot 1) \end{cases} \rightarrow A_1 = 0.0655, A_2 = -0.0655$$

$$i(t) = 4 + 0.0655(e^{-0.5218t} - e^{-11.978t})$$

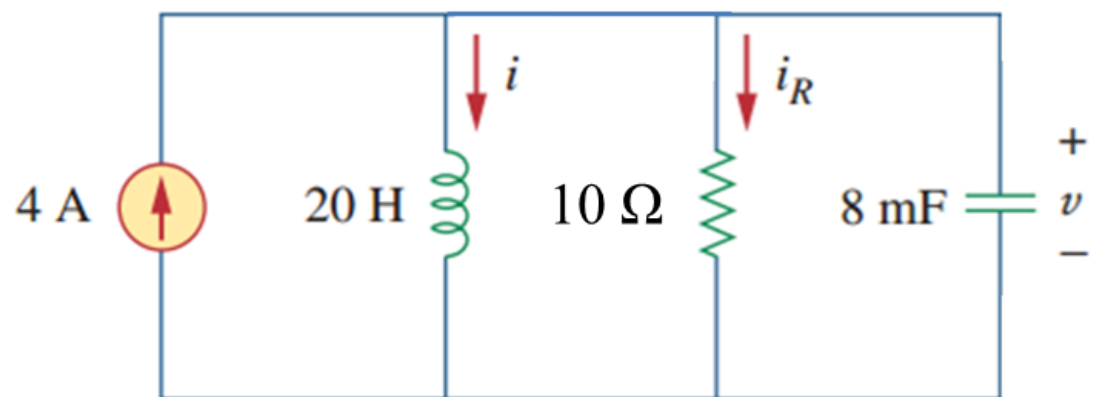
$$i_R(t) = \frac{v(t)}{20} = L \frac{di}{dt} \frac{1}{20} = 0.785e^{-11.978t} - 0.0342e^{-0.5218t} \text{ A}$$



## 8.8 Step response of parallel RLC circuits (h)

- Example 8.07

Solution: (5) Calculate other variables based on  $i(t)$ .



$$i(t) = 4 + 0.0655 \left( e^{-0.5218t} - e^{-11.978t} \right)$$

$$i_R(t) = \frac{v(t)}{20} = L \frac{di}{dt} \frac{1}{20}$$

$$= 0.785e^{-11.978t} - 0.0342e^{-0.5218t} \text{ A}$$

# Lesson 8 Second-order circuits

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|         |                       |              |
|---------|-----------------------|--------------|
| 8.1-8.3 | Second-order circuits | Slides 08-20 |
|---------|-----------------------|--------------|

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|         |                                  |              |
|---------|----------------------------------|--------------|
| 8.4-8.6 | Natural response of RLC circuits | Slides 22-50 |
|---------|----------------------------------|--------------|

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|         |                               |              |
|---------|-------------------------------|--------------|
| 8.7-8.8 | Step response of RLC circuits | Slides 52-70 |
|---------|-------------------------------|--------------|

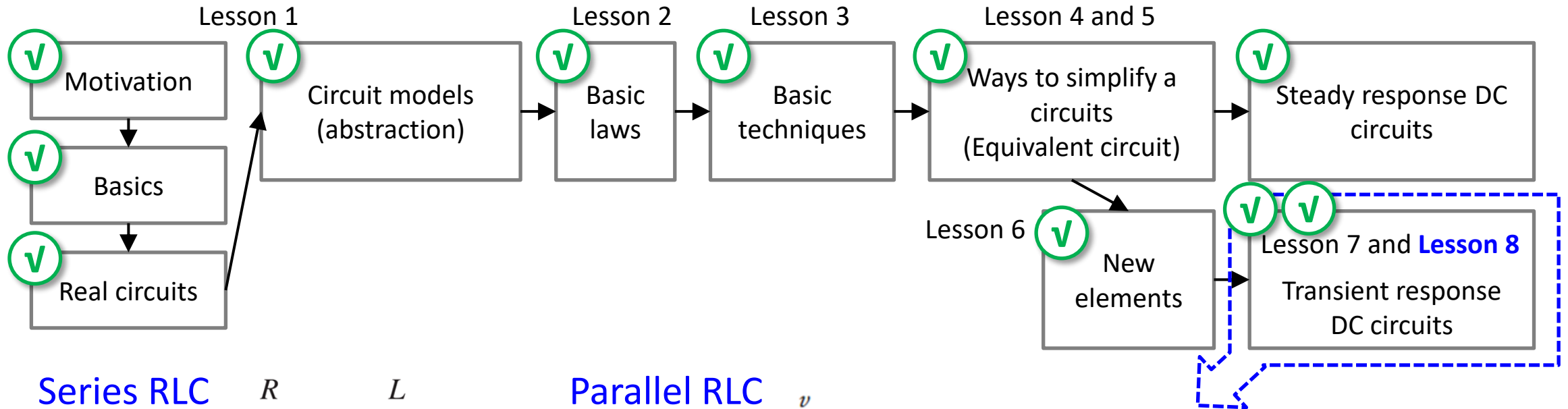
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|            |                |                     |
|------------|----------------|---------------------|
| <b>8.9</b> | <b>Summary</b> | <b>Slides 72-73</b> |
|------------|----------------|---------------------|

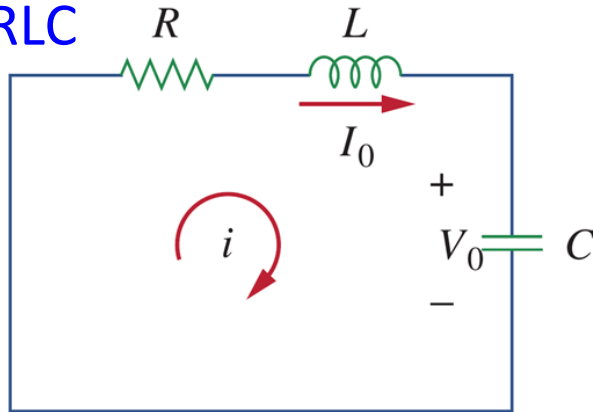
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## 8.9 Summary (a)

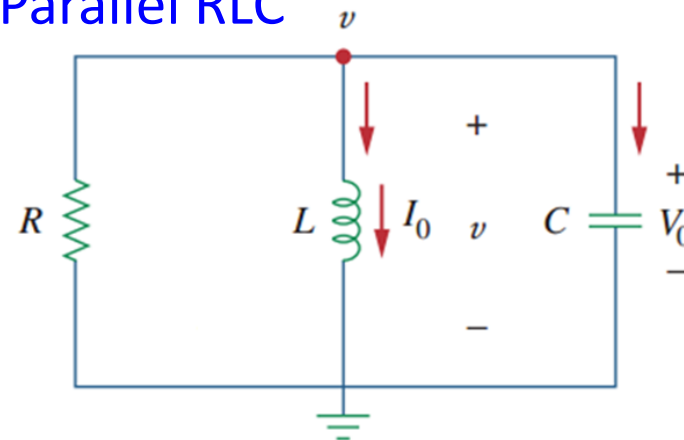
- EE104 road map



Series RLC



Parallel RLC



## 8.9 Summary (b)

While there are no assignments this week, these problems are recommended for practice:

Problems 8.4, 8.16, 8.22, 8.30, 8.50

- (a) A certain amount of practicing is necessary to sharpen your mind and to deepen your understanding.
- (b) There are plenty of problems in the textbook, try to complete as many as you can.

# Noctilucent Clouds

